

# Understanding the language of Cohomology (from a geometric point of view)

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# 1 What is (Co)homology? and Why Cohomology?

In mathematics, or even in other related fields such as topological data analysis, chain complexes are very popular. (Co)homology is a systematic technique by which we can extract information from these complexes and express them as homological invariants of rings, modules, topological spaces, etc.

Homological algebra was invented in the 19th century via the work of Riemann (1857) and Betti (1871) and later the study the homology groups of a space became more focused by the works of Emmy Noether in 1925. Most of 19th century consisted of a gradual effort by Riemann, Betti and Poincaré to define the Betti numbers of a piecewise linear manifold and other numerical topological invariants such as the Torsion coefficients of a topological space. Cohomology though being more fundamental to modern algebraic topology, its importance was not seen for some 40 years after the development of homology. Henri Poincaré used the concept of dual cell structure in his proof of Poincaré duality which marked the beginning of cohomology.

In brief homology theory could be described as: constructing a chain complex with given topological data, that is usually a sequence of abelian groups associated with a topological space and defining their homology groups to be the homology of that chain complex.

The construction of some versions of cohomology come by dualizing the above mentioned construction. One of the most fundamental aspects of studying cohomology is that it captures more richer algebraic invariants of a space than homology itself!

By now it is quite clear that there are many (co)homology theories. Some popular cohomology theories include sheaf cohomology, Algebraic K theory, Hochschild cohomology, galois cohomology, etc. but the one and one of the most important cohomology theories present, that is the main focus of this project is *De Rham cohomology* theory, which one would heuristically describe it to be a tool that measures how many different ways a closed differential form defined on a smooth manifold can fail to be exact.

## 2 Introduction: De Rham Cohomology

We begin by recalling the definitions of closed forms and exact forms. If  $M$  is a smooth  $n$  manifold, and  $\omega$  is a smooth  $k$  ( $k \leq n$ ) form defined on  $M$ , then  $\omega$  is said to be closed if  $d\omega = 0$  and it is said to be exact if there exists a smooth  $(k-1)$  form  $\lambda$  defined on  $M$  such that  $\omega = d\lambda$ .

The existence of exterior derivative  $d$  with the property  $d \circ d = 0$  tells us that every exact form is closed. But it is to question the converse: Is every closed form exact? is where De Rham cohomology comes in the picture. The answer to this question is not necessarily, as observed in the following example:

**Example 2.0.0.1.** Let us consider the smooth manifold  $\mathbb{R}^2 \setminus \{0\}$  and the following 1-form defined on it:

$$\omega = \frac{xdy - ydx}{x^2 + y^2}.$$

It is easy to check  $d\omega = 0$ , by defining  $\lambda = xdy - ydx$  and  $f(x, y) = \frac{1}{x^2 + y^2}$  and  $d\omega = d(f\lambda) = df \wedge \lambda + f(d\lambda)$ . Then, it's just a matter of easy computations that we would get  $d\omega = 0$ , hence  $\omega$  is closed. But,  $\omega$  is not exact. There are many ways to see it. One of them uses the well known proposition which states a smooth form defined on a smooth manifold is exact iff it is conservative. If we compute the integral of  $\omega$  about  $S^1$ , then the answer would come out to be  $2\pi \neq 0$ , which shows that  $\omega$  is not conservative, hence, not exact.

The problem that occurred in the above example is that there is a 'hole' in the center of the manifold. when we restrict the domain to the upper half plane ,i.e.  $\{(x, y) : y > 0\}$ , then  $\omega$  is exact and is equal to  $d\theta$ .

For higher degree closed forms, the exactness of these forms depends on the topological properties of the manifold, such as the existence of holes of higher dimensions. This dependence leads to a set of invariants of smooth manifolds, called the de Rham cohomology groups.

Later, we will prove that every closed form is locally exact and whether these closed forms are globally exact depends on the shape of the domain (we will prove that every closed form defined on a star shaped open subset of  $\mathbb{R}^n$  or  $\mathbb{H}^n$  is exact) and we will see how amazingly the de Rham cohomology groups relates to it.

**Remark 2.0.0.2.** If  $M$  is a smooth manifold and  $\omega$  is a smooth form defined on it, then, following are some other ways by which we can check whether  $\omega$  is exact:

1. By Stoke's Theorem, if  $\omega$  is an exact form, then the integral of  $\omega$  over every compact submanifold  $S$  of  $M$  without boundary is 0.
2.  $\omega$  is exact iff  $\omega$  is conservative.

So, now let us finally get to the point and define the de Rham cohomology groups:

### 3 The de Rham cohomology groups

Let  $M$  be a smooth manifold with or without boundary, of dimension  $n$ . The set of all differential  $p$ -forms form a real vector space  $\Omega^p(M)$ , which together with the exterior derivative  $d$  forms a cochain complex:

$$\dots \xrightarrow{d} \Omega^{p-1}(M) \xrightarrow{d} \Omega^p(M) \xrightarrow{d} \Omega^{p+1}(M) \xrightarrow{d} \dots$$

where  $p$  is a non-negative integer.

Since, the exterior derivative  $d$  is a  $\mathbb{R}$  linear map, its kernel and image are linear subspaces. So, we define the following:

$$\mathcal{Z}^p(M) := \text{Ker}\{d : \Omega^p(M) \rightarrow \Omega^{p+1}(M)\} = \{\text{closed } p\text{-forms on } M\}.$$

$$\mathcal{B}^p(M) := \text{Im}\{d : \Omega^{p-1}(M) \rightarrow \Omega^p(M)\} = \{\text{exact } p\text{-forms on } M\}.$$

Since, the exact forms are always closed we have that  $\mathcal{B}^p(M) \subseteq \mathcal{Z}^p(M)$ . Hence, the following quotient vector space is well defined:

$$H_{dR}^p(M) := \frac{\mathcal{Z}^p(M)}{\mathcal{B}^p(M)}. \quad (1)$$

This quotient space is called the  *$p$ -th de Rham group of  $M$  or de Rham cohomology group in degree  $p$  of  $M$* .

By group, we are here talking about the underlying group of this space under vector addition.

Here, we assume  $\Omega^p(M)$  to be the zero vector space if  $p < 0$  or  $p > n$ , so, it follows that  $\mathcal{Z}^n(M) = \Omega^n(M)$  and  $\mathcal{B}^0(M) = 0$ , and thus, by definition:

1.  $H_{dR}^p(M) = 0$  for  $p < 0$  or  $p > n$ .
2.  $H_{dR}^p(M) = 0$  for  $0 \leq p \leq n$  if and only if every closed  $p$ -form on  $M$  is exact.

**Remark 3.0.0.1.** In example 2.0.0.1, since the 1-form  $\omega$  is not exact, we have  $H_{dR}^1(\mathbb{R}^2 \setminus \{0\}) \neq 0$ .

Since, the de Rham cohomology group is afterall a quotient space, its elements are equivalence classes. For a smooth manifold  $M$ , the elements of  $H_{dR}^p(M)$  are equivalence classes  $[\omega]$ , called the *cohomology class* for  $\omega$ . Any two elements of a equivalence class differ by an exact form and are said to be *cohomologous*.

### 3.1 Induced Cohomology map

We know, if  $F : M \rightarrow N$  is a smooth map, then the pullback map  $F^*$  carries exact forms to exact forms and closed forms to closed forms, that is  $\forall 0 \leq p \leq n$ ,  $F^*$  carries  $\mathcal{Z}^p(N)$  to  $\mathcal{Z}^p(M)$  and  $\mathcal{B}^p(N)$  to  $\mathcal{B}^p(M)$ . So, by definition of de Rham cohomology groups, it is quite intuitive to think that  $F^*$  also carries  $H_{dR}^p(N)$  to  $H_{dR}^p(M)$ . And indeed it does!

We define the map as:

$$\begin{aligned} F^* : H_{dR}^p(N) &\longrightarrow H_{dR}^p(M) \\ [\omega] &\longmapsto [F^*\omega] \end{aligned}$$

We show that this map is well defined.

Consider  $\omega' = \omega + d\lambda$ . Then,  $[F^*\omega'] = [F^*(\omega + d\lambda)] = [F^*\omega + d(F^*\lambda)] = [F^*\omega]$ .

The following proposition follow directly from the definition, but it is quite fundamental, so we mention it:

**Proposition 3.1.0.1.** *i) If  $G : N \rightarrow P$  is another smooth map, then,*

$$(G \circ F)^* = F^* \circ G^* : H_{dR}^p(P) \longrightarrow H_{dR}^p(M).$$

*ii) If  $Id$  is the identity map of  $M$ , then,  $Id^*$  is the identity map of  $H_{dR}^p(M)$ .*

**Corollary 3.1.0.2. (Diffeomorphism invariance of de Rham Cohomology).** *Suppose  $M$  and  $N$  are two diffeomorphic manifolds (with or without boundary). Then, their de Rham cohomology groups are isomorphic.*

*Proof.* Let us suppose the manifolds have dimension  $n$ . Then, for an isomorphism  $F : M \rightarrow N$ , the pullback map  $F^* : H_{dR}^p(N) \rightarrow H_{dR}^p(M)$  is a vector space isomorphism for all  $0 \leq p \leq n$ .  $\square$

### 3.2 Some techniques to calculate some special de Rham groups

It is not always easy to calculate the cohomology groups directly, but in some special cases the following techniques help.

**Proposition 3.2.0.1.** *If  $M$  is a smooth manifold such that  $M = \Pi_j M_j$ , where each  $M_j$  is a smooth  $n$ -manifold, then for each  $p$ , the inclusion maps  $i : M_j \hookrightarrow M$  induce the following isomorphism:*

$$H_{dR}^p(M) \cong \Pi_j H_{dR}^p(M_j).$$

*Proof.* The pullback of these inclusion maps induce the following vector space isomorphism for each  $j$  and for each  $p$ :

$$\begin{aligned} i_j^* : \Omega^p(M) &\longrightarrow \Omega^p(M_j) \\ \omega &\longmapsto i_j^*\omega \end{aligned}$$

Hence, we have our required vector space isomorphism:

$$\begin{aligned} \eta : \Omega^p(M) &\longrightarrow \Pi_j \Omega^p(M_j) \\ \omega &\longmapsto (i_1^*\omega, i_2^*\omega, \dots) = (\omega|_{M_1}, \omega|_{M_2}, \dots) \end{aligned}$$

For any  $p$ -form  $\omega$ , if the restriction to each component  $M_j$  is 0, then,  $\omega$  is itself zero. Hence,  $\eta$  is injective. Also, if we define an arbitrary  $p$ -form on each of the manifolds  $M_j$ , then it clearly defines a form on the manifold  $M$ . Hence,  $\eta$  is surjective. Thus,  $\eta$  is an isomorphism.  $\square$

From the above proposition, it is clear that it is enough to work with connected components of the manifold, as the de Rham cohomology group of the entire manifold  $M$ , for each  $p$ , will be the direct product of the de Rham cohomology groups of the connected components of  $M$ .

**Proposition 3.2.0.2. (Cohomology in degree zero).** *The de Rham cohomology group of a smooth connected manifold in degree 0 is a space consisting of all constant functions and is therefore 1-dimensional.*

*Proof.* The 0 forms are the set of all smooth functions defined on the manifold  $M$  and thus the space  $Z^0(M)$  consists of  $f \in C^\infty(M)$  such that  $df = 0$ . And, since,  $M$  is a connected manifold,  $f$  satisfies such a property iff  $f$  is constant on  $M$ . We have already defined  $\mathcal{B}^0(M)$  to be 0. Hence, we have,

$$H_{dR}^0(M) = \mathcal{Z}^0(M) = \{\text{constant functions defined on } M\}.$$

□

**Remark 3.2.0.3.** If we consider a disconnected zero manifold, then, it is just a discrete set of points. And thus by above proposition,  $H_{dR}^0(M)$  is the direct product of 1-dimensional vector spaces. And, the other higher degree cohomology groups are 0.

## 4 de Rham groups as invariants under homotopy

As we saw in proposition 3.1.0.2, any two diffeomorphic manifolds have isomorphic de Rham cohomology groups. But, these de Rham groups are invariants under much more stronger settings, such as, not only just topological<sup>1</sup>, but they are in fact homotopy invariants!

To show this, we first prove that for any two smooth maps  $F, G : M \rightarrow N$  which are homotopically equivalent, their induced cohomology maps  $F^*, G^* : H_{dR}^p(N) \rightarrow H_{dR}^p(M)$  are equal, that is, for all  $[\omega] \in H_{dR}^p(N)$  we have  $F^*[\omega] = G^*[\omega]$ .

We choose an arbitrary closed  $p$ -form  $\omega$  on  $N$ , and we construct a  $(p-1)$  form  $\lambda$  on  $N$  such that:

$$F^*\omega - G^*\omega = d\lambda$$

from which it will follow,

$$F^*[\omega] - G^*[\omega] = [d\lambda] = 0. \quad (2)$$

To construct such a  $\lambda$ , we introduce a family of linear maps  $\{h_p : \Omega^p(N) \rightarrow \Omega^{p-1}(M)\}$ , which for each  $\omega$  satisfies the following:

$$d(h_p\omega) + h_{p+1}(d\omega) = G^*\omega - F^*\omega. \quad (3)$$

Such a collection of linear operators  $\{h_p\}$  are called *cochain homotopy operators between  $F^*$  and  $G^*$* .

Clearly, (3) implies (2) when  $\omega$  is closed.

The following diagram will help in understanding the operators  $h_p$  articulately (in the diagram  $F^*$  and  $G^*$  are denoted as usual pullbacks, but in this discussion we are referring to induced cohomology maps):

$$\begin{array}{ccccccc} \dots & \longrightarrow & \Omega^{p-1}(M) & \longrightarrow & \Omega^p(M) & \longrightarrow & \Omega^{p+1}(M) \longrightarrow \dots \\ & & \nwarrow h_p & & \uparrow F^* & \uparrow G^* & \nwarrow h_{p+1} \\ \dots & \longrightarrow & \Omega^{p-1}(N) & \longrightarrow & \Omega^p(N) & \longrightarrow & \Omega^{p+1}(N) \longrightarrow \dots \end{array}$$

**Proposition 4.0.0.1.** *Let  $M$  and  $N$  be smooth manifolds with or without boundary and  $F, G : M \rightarrow N$  are smooth maps such that there exists homotopy operator between the pullback maps  $F^*, G^*$ , then the induced cohomology maps  $F^*, G^* : H_{dR}^p(N) \rightarrow H_{dR}^p(M)$  are equal.*

To prove the homotopy invariance, we consider the following map:

For a smooth manifold  $M$  and for each  $t \in I$  we have:

$$\begin{aligned} i_t : M &\longrightarrow M \times I \\ x &\longmapsto (x, t) \end{aligned}$$

With these above defined map, we have the following lemma:

<sup>1</sup>Though after proving the homotopy invariance, the de Rham groups being topological invariants is a trivial fact, but it is still worth keeping a note of it because the construction of de Rham groups being topological, it nowhere assures that if a topological manifold is given different differentiable structures, the cohomology groups still remain the same.

**Lemma 4.0.0.2. (Existence of Homotopy operator).** *For any smooth manifold  $M$  with or without boundary, there exists a homotopy operator between the two pullback maps  $i_0^*, i_1^* : \Omega^*(M \times I) \rightarrow \Omega^*(M)$ .*

*Proof.* As per definition, to prove the existence of such a homotopy operator between the pullback maps  $i_0^*$  and  $i_1^*$ , for each  $p$ , we need to define a linear map  $h : \Omega^p(M \times I) \rightarrow \Omega^{p-1}(M)$  such that it satisfies

$$h(d\omega) + d(h\omega) = i_1^*\omega - i_0^*\omega \quad (4)$$

for all  $\omega \in \Omega^p(M \times I)$ . ( We have dropped the suffix  $p$  in the notation of homotopy operator  $h$  for simplification.)

$$\begin{array}{ccccccc} \dots & \longrightarrow & \Omega^{p-1}(M) & \longrightarrow & \Omega^p(M) & \longrightarrow & \Omega^{p+1}(M) \longrightarrow \dots \\ & & \nwarrow h & & \uparrow i_0^* \uparrow i_1^* & & \nwarrow h \\ \dots & \longrightarrow & \Omega^{p-1}(M \times I) & \longrightarrow & \Omega^p(M \times I) & \longrightarrow & \Omega^{p+1}(M \times I) \longrightarrow \dots \end{array}$$

Let  $s$  denote the standard coordinate on  $\mathbb{R}$  and let  $q \in M$  be arbitrary. Let us consider the vector field  $S$  on  $M \times \mathbb{R}$  given by  $S_{(q,s)} = (0, \frac{\partial}{\partial s}|_s)$  under the usual identification of tangent spaces  $T_{(q,s)}(M \times \mathbb{R}) \leftrightarrow T_q M \times T_s \mathbb{R}$ . Note that the vector field  $S$  is tangent to the boundary of the manifold  $M \times \mathbb{R}$ , which is the reason behind the choice of such a vector field, as in our course of calculations for defining such a homotopy operator, we need to consider the Lie derivative of a  $p$ -form along the vector field  $S$ , and we know that in manifolds with boundary such Lie derivatives exist and are smooth only if the vector field is tangent to the boundary of the manifold. (Refer to remark A.2.2.1). Now, let us define the homotopy operator  $h$ :

Given a smooth  $p$ -form  $\omega \in \Omega^p(M \times I)$ , we define  $h\omega \in \Omega^{p-1}(M)$  by

$$h\omega = \int_0^1 i_t^*(S \lrcorner \omega) dt, \quad (5)$$

that is, in particular,

$$(h\omega)_q = \int_0^1 i_t^*((S \lrcorner \omega)_{(q,t)}) dt.$$

The above integrand is a function of  $q$  and  $t$  taking value in the vector space  $\Lambda^{p-1}(T_q^* M)$ , which is a topological vector space, hence the integral makes sense. Let  $U$  be a smooth chart of  $M$  containing the point  $q$ .

Let us define

$$\begin{aligned} \alpha : U \times I &\longrightarrow \Lambda^{p-1}(T_q^* M) \\ (q, t) &\longmapsto i_t^*(S \lrcorner \omega). \end{aligned}$$

In local coordinates,  $\alpha(q, t) = \sum_I \alpha_I(q, t) dx_I$ , for all  $(q, t) \in U \times I$ , where  $I = (i_1 < i_2 < \dots < i_{p-1})$  is an increasing sequence of  $(p-1)$  indices. Now, these coefficient functions  $\alpha_I$  are smooth functions of  $(q, t) \in U \times I$ , and since  $q$  is arbitrary, the integral defines a smooth  $(p-1)$ -form on  $M$ .

Now we claim that

$$d(h\omega) = d\left(\int_0^1 i_t^*((S \lrcorner \omega)_{(q,t)}) dt\right) = \int_0^1 d(i_t^*(S \lrcorner \omega)_{(q,t)}) dt. \quad (6)$$

Suppose the above claim is true. Also, from equation (5) we have

$$h(d\omega) = \int_0^1 i_t^*(S \lrcorner d\omega) dt. \quad (7)$$

Now, adding equations (6) and (7) and using Cartan's magic formula, which states

$$\mathcal{L}_S \omega = S \lrcorner d\omega + d(S \lrcorner \omega)$$

where  $\mathcal{L}_S \omega$  is the lie derivative of the  $p$ -form  $\omega$  along the vector field  $S$ , we get

$$\begin{aligned} h(d\omega) + d(h\omega) &= \int_0^1 i_t^*(S \lrcorner d\omega) + d(i_t^*(S \lrcorner \omega)) dt \\ &= \int_0^1 i_t^*(S \lrcorner d\omega) + i_t^*(d(S \lrcorner \omega)) dt \\ &= \int_0^1 i_t^*(S \lrcorner d\omega + d(S \lrcorner \omega)) dt \\ &= \int_0^1 i_t^*(\mathcal{L}_S \omega) dt. \end{aligned}$$

To show (4), by fundamental theorem of calculus and the above integral, it follows that we reduce down to proving

$$i_t^*(\mathcal{L}_S \omega) = \frac{d}{dt} i_t^* \omega. \quad (8)$$

Here,  $\mathcal{L}_S \omega$  exists and is smooth as discussed before. Now for proving this, we consider a flow  $\theta_t$  of  $S$  on  $M \times \mathbb{R}$ . Fundamental theorem of flows (A.2.2.1) assures the existence of such a flow. By our choice of  $S$ , we observe that  $\theta_t(q, s) = (q, t + s)$ , which implies that  $S$  has a global flow. Henceforth,  $S$  is a complete vector field. It follows that we can write  $i_t = \theta_t \circ i_0$ . Thus,

$$\begin{aligned} i_t^*(\mathcal{L}_S \omega) &= i_0^* \circ \theta_t^*(\mathcal{L}_S \omega) \\ &= i_0^* \left( \frac{d}{dt} \theta_t^* \omega \right) \quad (\text{by A.2.2.1}) \\ &= \frac{d}{dt} (i_0^* \circ \theta_t^*) \omega \\ &= \frac{d}{dt} i_t^* \omega, \end{aligned}$$

Now, let us prove our claim, equation (6).

We already have  $i_t^*(S \lrcorner \omega) = \sum_I \alpha_I(q, t) dx_I$ . Then,  $\int_0^1 i_t^*(S \lrcorner \omega) = \sum_I \int_0^1 \alpha_I(q, t) dt dx_I$ . Let,  $\int_0^1 \alpha_I(q, t) dt = \beta_I$ . It follows that  $\int_0^1 i_t^*(S \lrcorner \omega) = \sum_I \beta_I(q) dx_I$ . Thus,  $d(\int_0^1 i_t^*(S \lrcorner \omega)) = d(\sum_I \beta_I dx_I) = \sum_I d\beta_I \wedge dx_I$ .

On the other hand,  $d(i_t^*(S \lrcorner \omega)) = d(\sum_I \alpha_I(q, t) dx_I) = \sum_I d\alpha_I(q, t) \wedge dx_I$ . It follows that,  $\int_0^1 d(i_t^*(S \lrcorner \omega)) dt = \sum_I \int_0^1 d\alpha_I(q, t) \wedge dx_I dt = \sum_I d\beta_I \wedge dx_I$ .  $\square$

**Remark 4.0.0.3.** From proposition 4.0.0.1 and lemma 4.0.0.2, we have that the induced cohomology maps  $i_0^*, i_1^* : H_{dR}^p(M \times I) \rightarrow M$  are equal.

**Proposition 4.0.0.4.** Let  $M$  and  $N$  be smooth manifolds with or without boundary and let  $F, G : M \rightarrow N$  be two homotopic smooth maps. Then, the induced cohomology maps  $F^*, G^* : H_{dR}^p(N) \rightarrow H_{dR}^p(M)$  are equal.

*Proof.* Since,  $F$  and  $G$  are smooth maps that are homotopically equivalent, we have that there exists a smooth map  $H : M \times I \rightarrow N$  such that  $H(x, 0) = F(x)$  and  $H(x, 1) = G(x)$  for all  $x \in M$ .

$$\begin{array}{ccccc} & & G=H \circ i_1 & & \\ & \nearrow & & \searrow & \\ M & \xrightarrow{i_1} & M \times I & \xrightarrow{H} & N \\ & \searrow & & \nearrow & \\ & & F=H \circ i_0 & & \end{array}$$

From the above diagram, proposition 3.1.0.1 and remark 4.0.0.3, it follows that:

$$F^* = (H \circ i_0)^* = i_0^* \circ H^* = i_1^* \circ H^* = (H \circ i_1)^* = G^*.$$

$\square$

Now, we finally prove the invariance of de Rham cohomology groups of homotopically equivalent manifolds:



**Theorem 4.0.0.5.** *If  $M$  and  $N$  are homotopically equivalent smooth manifolds with or without boundary, then,*

$$H_{dR}^p(M) \cong H_{dR}^p(N).$$

*Proof.* Since,  $M$  and  $N$  are homotopically equivalent smooth manifolds, there exist a continuous map  $F : M \rightarrow N$  which is a homotopy equivalence, that is, there exists a continuous map  $G : N \rightarrow M$  such that  $G \circ F \simeq id_M$  and  $F \circ G \simeq id_N$ . By Whitney Approximation theorem, we know any continuous map can be approximated by a smooth map. Hence, there exists smooth maps  $\tilde{F} : M \rightarrow N$  and  $\tilde{G} : N \rightarrow M$  such that  $\tilde{G} \circ \tilde{F} \simeq id_M$  and  $\tilde{F} \circ \tilde{G} \simeq id_N$  (as composition preserves homotopy).

Now to show that  $H_{dR}^p(M) \cong H_{dR}^p(N)$ , we consider the induced cohomology maps  $\tilde{F}^* : H_{dR}^p(N) \rightarrow H_{dR}^p(M)$  and  $\tilde{G}^* : H_{dR}^p(M) \rightarrow H_{dR}^p(N)$  and we have that  $\tilde{G}^* \circ \tilde{F}^* = (\tilde{F} \circ \tilde{G})^* = (id_N)^* = id_{H_{dR}^p(N)}$  and similarly,  $\tilde{F}^* \circ \tilde{G}^* = id_{H_{dR}^p(M)}$ , so both the cohomology maps are isomorphisms.  $\square$

## 4.1 Further techniques to compute de Rham groups using homotopy invariance

After that we have proved the homotopy invariance of de Rham groups, one obvious thought that one would get is about the de Rham group of contractible<sup>2</sup> manifolds.

**Theorem 4.1.0.1. (de Rham groups of contractible manifolds).** *If  $M$  is a contractible smooth manifold, then  $H_{dR}^p(M) = 0$ , for all  $p \geq 1$ .*

*Proof.* Since,  $M$  is contractible, there exists a point  $q \in M$  such that the identity map  $id_M \simeq c_q$ , where  $c_q : M \rightarrow M$ , is a map that maps the entire of  $M$  to the point  $q$ . Now, considering the inclusion map  $i_q : \{q\} \rightarrow M$ , we have  $i_q \circ c_q \simeq id_M$  and  $c_q \circ i_q \simeq id_q$ . Hence, the constant map  $c_q$  is a homotopy equivalence. And, since a point is a 0-dimensional manifold, clearly, for all  $p \geq 1$ , we have  $H_{dR}^p(\{q\}) = 0$ , and, therefore,  $H_{dR}^p(M) = 0$ , for all  $p \geq 1$ .  $\square$

Here are some results that we have previously talked about and will now finally prove using homotopy invariance:

### 4.1.1 The Poincaré lemma

**Definition 4.1.1.1. (Star shaped domain).** A subset  $U \subseteq \mathbb{R}^n$  is said to be star shaped if  $\exists c \in U$  such that for any point  $x \in U$ , there exists a straight line segment joining  $c$  and  $x$  that is totally contained in  $U$ .

**Theorem 4.1.1.2.** *If  $U$  is a star shaped open subset of  $\mathbb{R}^n$  or  $\mathbb{H}^n$ , then  $H_{dR}^p(U) = 0$  for all  $p \geq 1$ .*

*Proof.* Suppose  $U$  is star shaped with respect to a point  $c \in U$ . We consider the following straight line homotopy:

$$H(x, t) = c + t(x - c).$$

Clearly,  $H(x, 0) = c$  and  $H(x, 1) = x$ , that is the identity map of the subset  $U$  is homotopically equivalent to the constant map at  $c$ . Hence,  $U$  is contractible, thus,  $H_{dR}^p(U) = 0$ .  $\square$

**Remark 4.1.1.3.** The above theorem says that every star shaped open subset of  $\mathbb{R}^n$  or  $\mathbb{H}^n$  is contractible.

**Example 4.1.1.4.** Consider the open unit disk  $\mathbb{B}^n = \{x \in \mathbb{R}^n : \|x\| < 1\}$ . It is star shaped (infact convex) open subset of  $\mathbb{R}^n$ , hence contractible. Thus,  $H_{dR}^p(\mathbb{B}^n) = 0$  for all  $p \geq 1$ .

**Remark 4.1.1.5.** Instead of  $\mathbb{B}^n$  in the above example if we consider  $\mathbb{R}^n$  or  $\mathbb{H}^n$ , then the same holds, hence, **cohomology groups of Euclidean spaces and half spaces are 0.**

**Corollary 4.1.1.6. (Local exactness of closed forms).** *Every closed form defined on a smooth manifold is locally exact.*

*Proof.* Let  $M$  be a smooth manifold. For any arbitrary point  $q \in M$ , there is a neighbourhood  $U$  of this point  $q$  in  $M$  which is diffeomorphic to an open connected subset of  $\mathbb{R}^n$ . Since, such subsets are always convex, hence star shaped, from diffeomorphism invariance of cohomology groups, it follows that  $H_{dR}^p(U) = 0$ . Thus, it implies that closed forms are exact in this open subset  $U$ . Hence, if  $\omega$  is a closed form defined on  $M$ , then, each point of  $M$  has neighbourhood in which  $\omega$  is exact. Therefore, closed forms are locally exact.  $\square$

<sup>2</sup>a topological space  $X$  is said to be contractible if the identity map  $id_X$  is homotopic to a constant map.

## 5 Relation between first cohomology and first homotopy group

Let  $M$  be a connected smooth manifold and let  $q$  be an arbitrary point in  $M$ .

Consider the following  $\mathbb{R}$ -vector space:

$$\text{Hom}(\pi_1(M, q), \mathbb{R}) = \{\eta : \pi_1(M, q) \rightarrow \mathbb{R} \mid \eta \text{ is a group homomorphism}\}$$

where  $\pi_1(M, q)$  is the first homotopy group or the fundamental group of the manifold  $M$  based at the point  $q$ .

$\text{Hom}(\pi_1(M, q), \mathbb{R})$  forms a vector space under pointwise addition of homomorphisms and multiplication by constants.

We define the following  $\mathbb{R}$ -linear map:

$$\begin{aligned} \Phi : H_{dR}^1(M) &\longrightarrow \text{Hom}(\pi_1(M, q), \mathbb{R}) \\ [\omega] &\longmapsto \Phi[\omega] \end{aligned}$$

where the group homomorphism  $\Phi[\omega]$  is defined as follows:

$$\begin{aligned} \Phi[\omega] : \pi_1(M, q) &\longrightarrow \mathbb{R} \\ [\gamma] &\longmapsto \int_{\tilde{\gamma}} \omega \end{aligned}$$

where  $[\gamma]$  is a path homotopy class in  $\pi_1(M, q)$  and  $\tilde{\gamma}$  is a piecewise smooth curve as a representation of the same path class  $[\omega]$ .

The map  $\Phi$  is actually an isomorphism, but since the proof is a bit non-trivial with the tools we have right now, we just show that for each  $q$ , the map  $\Phi$  is well defined and injective.

To begin with, we first mention the following lemma:

**Lemma 5.0.0.1.** *If  $\omega$  is a smooth closed 1-form defined on a smooth manifold  $M$ , then*

$$\int_{\gamma_0} \omega = \int_{\gamma_1} \omega$$

where  $\gamma_0$  and  $\gamma_1$  are two homotopic smooth curve segments.

**The map  $\Phi[\omega]$  is well defined for some  $\omega$  in  $H_{dR}^p(M)$ :**

Suppose  $\gamma_0$  and  $\gamma_1$  are two homotopic loops in  $\pi_1(M, q)$  such that  $[\gamma_0] = [\gamma_1]$ . Then, by Whitney approximation, we have two smooth homotopic loops  $\tilde{\gamma}_0$  and  $\tilde{\gamma}_1$  in the same path homotopy classes of  $\gamma_0$  and  $\gamma_1$  respectively, and hence both of their equivalence classes are equal. By lemma 5.0.0.1, we have that

$$\int_{\tilde{\gamma}_0} \omega = \int_{\tilde{\gamma}_1} \omega$$

**$\Phi[\omega]$  is a group homomorphism:**

Let  $[\gamma_1], [\gamma_2] \in \pi_1(M, q)$ . We know,  $[\gamma_1] \cdot [\gamma_2] = [\gamma_1 \cdot \gamma_2]$

Now,

$$\Phi[\omega][\gamma_1 \cdot \gamma_2] = \int_{\tilde{\gamma}_1 \tilde{\gamma}_2} \omega = \int_{\tilde{\gamma}_1} \omega + \int_{\tilde{\gamma}_2} \omega = \Phi[\omega][\gamma_1] + \Phi[\omega][\gamma_2].$$

**Well definedness of the map  $\Phi$ :**

Consider  $\omega'$  such that  $[\omega] = [\omega']$ . Hence,  $\exists$  a  $f \in C^\infty(M)$  such that  $\omega' = \omega + df$ .

Now,

$$\Phi[\omega'][\tilde{\gamma}] = \int_{\tilde{\gamma}} \omega' = \int_{\tilde{\gamma}} (\omega + df) = \int_{\tilde{\gamma}} \omega + \int_{\tilde{\gamma}} df = \int_{\tilde{\gamma}} \omega = \Phi[\omega][\tilde{\gamma}]$$

as integral of an exact form over a closed path is 0.

**Linearity of the map  $\Phi$ :** It follows from the linearity of the integral  $\int_{\tilde{\gamma}} \omega$ .

**Injectivity of the map  $\Phi$ :**

We first prove that if  $\int_{\tilde{\gamma}} \omega = 0$  for all  $\tilde{\gamma}$  then  $\omega$  is exact, where  $\tilde{\gamma}$  is piecewise smooth closed curve based at  $q$ . It is sufficient to show that  $\omega$  is conservative by remark 2.0.0.2.2.

Let suppose  $\sigma$  is another closed curve based at some point  $q' \neq q$ . Consider a piecewise smooth path  $\alpha$  from  $q$  to  $q'$ . Then,  $\alpha \cdot \sigma \cdot \bar{\alpha}$  is a piecewise smooth closed curve based at  $q$ . Then,

$$0 = \int_{\alpha \cdot \sigma \cdot \bar{\alpha}} \omega = \int_{\alpha} \omega + \int_{\sigma} \omega - \int_{\bar{\alpha}} \omega = \int_{\sigma} \omega.$$

Hence,  $\omega$  is conservative, thus, exact.

$$\begin{aligned} \text{Ker } \Phi &= \{[\omega] \in H_{dR}^1(M) : \Phi[\omega][\tilde{\gamma}] = 0, \text{ for all } [\tilde{\gamma}] \in \pi_1(M, q)\} \\ &= \{[\omega] \in H_{dR}^1(M) : \int_{\tilde{\gamma}} \omega = 0\} \\ &= \{[\omega] \in H_{dR}^p(M) : \omega \text{ is exact}\} \\ &= \{0\}. \end{aligned}$$

## 5.1 Some interesting results

The following lemma is a nice result obtained from the above monomorphism:

**Lemma 5.1.0.1.** *On a simply connected smooth manifold, every closed 1-form is exact.*

*Proof.* Let,  $M$  be a smooth manifold which is simply connected. We know for simply connected topological spaces, fundamental group is always 0. Hence,  $\pi_1(M, q) = 0$ . Since,  $\Phi$  is injective, we have that  $H_{dR}^1(M) = 0$ . Thus, all closed 1-forms are exact on  $M$ .  $\square$

**Remark 5.1.0.2.** The above lemma can also be proved in the following way:

Since  $M$  is simply connected, every piecewise smooth closed curve segment on  $M$  is null homotopic, that is, it is path homotopic to a constant curve. By lemma 5.0.0.1, integral of any closed 1-form over any closed curve segment is 0, hence every closed 1-form is conservative on  $M$ . It follows that every closed 1-form is exact on  $M$ .

The following corollary is a generalization of the above lemma:

**Corollary 5.1.0.3.** *If  $M$  is a connected smooth manifold, such that for each  $q \in M$ , we have  $\pi_1(M, q) < \infty$ , then,  $H_{dR}^1(M) = 0$ .*

*Proof.* Since, the fundamental group of  $M$  is finite, there cannot exist any non trivial group homomorphism from  $\pi_1(M, q)$  to  $\mathbb{R}$ . Hence,  $\text{Hom}(\pi_1(M, q), \mathbb{R}) = 0$ . Again, since,  $\Phi$  is injective, we have  $H_{dR}^1(M) = 0$ .  $\square$

**Remark 5.1.0.4.** The above corollary can be more generalized to manifolds having fundamental groups with each element of the group having finite order. It is because the non-existence of any non-trivial group homomorphism from a group, say,  $\Gamma$  to  $\mathbb{R}$  doesn't depend upon the finiteness of the group, but, on the elements of the group having finite order.

Thus, if  $\Gamma$  is a torsion group, then, every element of  $\Gamma$  has finite order. So, **manifolds having torsion groups as their fundamental groups have de Rham cohomology group of degree 1 as 0.**

**Example 5.1.0.5.** Consider the  $n$ -sphere,  $S^n = \{x \in \mathbb{R}^n : \|x\| = 1\}$ . By using the above results and the previously stated monomorphism, we can compute the de Rham cohomology groups on  $S^n$  in degree 1 very easily.

We show

$$H_{dR}^1(S^n) \cong \begin{cases} \mathbb{R} & \text{if } n \leq 1 \\ 0 & \text{if } n \geq 2 \end{cases}$$

By proposition 3.2.0.2, we have  $H_{dR}^0(S^n)$  is 1-dimensional, hence isomorphic to  $\mathbb{R}$ . The main chunk is to prove  $H_{dR}^1(S^1) \cong \mathbb{R}$ , for which it suffices to show that none of the orientation forms of  $S^1$  is exact, that is no top-degree closed form defined over  $S^1$  is exact. Let  $\omega$  be an orientation form on  $S^1$ . Then,  $\omega \neq 0$ , which implies  $\omega$  is either  $> 0$  or  $< 0$ . Since, tangent spaces to  $S^1$  are one-dimensional, it follows that  $\omega = f(t)dt$  for some  $f \in C^\infty(S^1)$ , where  $t \in (0, 2\pi)$ . It follows that

$$\int_{S^1} \omega = \int_0^{2\pi} f(t)dt$$

which is clearly non-zero. Thus,  $\omega$  is not conservative, and hence not exact, which implies  $\dim H_{dR}^1(S^1) \geq 1$ .

We know,  $\pi_1(S^1, 1) = \mathbb{Z}$ . And, it is an easy check that  $\text{Hom}(\pi_1(S^1, 1), \mathbb{R})$ , that is,  $\text{Hom}(\mathbb{Z}, \mathbb{R})$  is 1-dimensional. From the injectivity of the above monomorphism  $\Phi$ , it follows that  $H_{dR}^1(S^1)$  has dimension  $\leq 1$ . Therefore,  $\dim H_{dR}^1(S^1) = 1$  and so we have,  $H_{dR}^1(S^1) \cong \mathbb{R}$ .

Also, we know,  $S^n$  for  $n \geq 2$  is simply connected. Hence, by lemma 5.1.0.1, we have  $H_{dR}^1(S^n) = 0$ , for all  $n \geq 2$ . The result follows.

## 6 The Mayer-Vietoris Theorem

We have already seen that computations of de Rham cohomology groups of any smooth manifold  $M$  reduces down to computation of cohomology groups of its connected components by proposition 3.2.0.2. In some cases, we can further reduce down to computing the de Rham cohomology groups of open submanifolds of  $M$  with simpler cohomology, where  $M$  is the union of these open submanifolds.

This is where the *Mayer-Vietoris sequence* comes in the picture. It is a natural long exact sequence consisting of cohomology groups of the entire manifold, the direct sum of the cohomology groups of the open submanifolds and the cohomology groups of the intersection of these submanifolds. This sequence helps us to compute or at least provide us with some partial information about the cohomology groups in very difficult scenarios.

But in our case, we can easily compute the de Rham cohomology groups of spheres, punctured Euclidean spaces and also, the top degree cohomology groups of compact manifolds using the Mayer-Vietoris sequence.

*Setup of the theorem:*

Let  $M$  be a smooth manifold and  $U$  and  $V$  be its open subsets such that

$$M = U \cup V.$$

Consider the following four inclusions

$$\begin{array}{ccccc} & & U & & \\ & \nearrow i & & \searrow k & \\ U \cap V & & & & M \\ & \searrow j & & \nearrow l & \\ & & V & & \end{array}$$

which induce the following pullback maps:

$$\begin{array}{ccc}
 & \Omega^p(U) & \\
 i^* \swarrow & & \nwarrow k^* \\
 \Omega^p(U \cap V) & & \Omega^p(M) \\
 j^* \swarrow & & \nwarrow l^* \\
 & \Omega^p(V) &
 \end{array}$$

Since,  $U$  and  $V$  are open submanifolds of  $M$ , the above pullback maps are just restriction of the forms to the vectors tangent to the corresponding submanifolds. For example, for  $\omega \in \Omega^p(M)$ ,  $k^*\omega$  is just  $\omega|_U$ . Also, since the pullbacks commute with  $d$ , these pullback maps result in the corresponding induced cohomology maps.

Consider the following sequence of maps:

$$0 \longrightarrow \Omega^p(M) \xrightarrow{k^* \oplus l^*} \Omega^p(U) \oplus \Omega^p(V) \xrightarrow{i^* - j^*} \Omega^p(U \cap V) \longrightarrow 0$$

where,  $(k^* \oplus l^*)\omega = (k^*\omega, l^*\omega)$  and  $(i^* - j^*)(\omega, \eta) = i^*\omega - j^*\eta$ .

Now we finally state the Mayer-Vietoris Theorem:

**Theorem 6.0.0.1.** *Let  $M$  be a smooth manifold with or without boundary and let  $U, V$  be open in  $M$  such that  $M = U \cup V$ . Then, for each  $p$ , there is a linear map  $\delta : H_{dR}^p(U \cap V) \longrightarrow H_{dR}^{p+1}(M)$  such that the following sequence called the **Mayer-Vietoris sequence** is exact:*

$$\dots \xrightarrow{\delta} H_{dR}^p(M) \xrightarrow{k^* \oplus l^*} H_{dR}^p(U) \oplus H_{dR}^p(V) \xrightarrow{i^* - j^*} H_{dR}^p(U \cap V) \xrightarrow{\delta} H_{dR}^{p+1}(M) \xrightarrow{k^* \oplus l^*} \dots$$

## 7 Applications of Mayer-Vietoris Theorem

We will now use the Mayer-Vietoris theorem to easily calculate all the de Rham groups of  $n$ -sphere and punctured Euclidean plane.

### 7.1 Cohomology of $n$ -Spheres

**Theorem 7.1.0.1.** *For  $n \geq 1$ , the de Rham cohomology groups of  $S^n$  are given by*

$$H_{dR}^p(S^n) \cong \begin{cases} \mathbb{R} & \text{if } p = 0 \text{ or } p = n \\ 0 & \text{if } 0 < p < n \end{cases}$$

*Proof.* We prove this by induction. The case  $n = 0$  and  $n = 1$  has already been proved in example 5.1.0.5. Now, let us assume that the theorem is true for  $S^{n-1}$ , that is

$$H_{dR}^p(S^{n-1}) \cong \begin{cases} \mathbb{R} & \text{if } p = n - 1 \\ 0 & \text{if } 0 < p < n - 1 \end{cases}$$

To show that it is true for  $S^n$  also, it is enough to prove the following claim:

**Claim:**  $H_{dR}^p(S^n) \cong H_{dR}^{p-1}(S^{n-1})$ .

This above claim assures that the degree  $p$  cohomology groups of  $S^n$  can be computed just by knowing the degree  $p - 1$  cohomology groups (and thus indeed, any higher degree ( $\geq 2$ ) cohomology group of  $S^n$  reduces down to computing the cohomology group of  $S^n$  of degree 1, which we already know).

Now, we prove the claim using Mayer-Vietoris sequence.

Let  $U = S^n \setminus \{N\}$ ,  $V = S^n \setminus \{S\}$ . Then,  $S^n = U \cup V$ . By Mayer-Vietoris Theorem, there exists a  $\mathbb{R}$ -linear map  $\delta : H_{dR}^{p-1}(U \cap V) \rightarrow H_{dR}^p(S^n)$  such that the following sequence is exact:

$$H_{dR}^{p-1}(U) \oplus H_{dR}^{p-1}(V) \longrightarrow H_{dR}^{p-1}(U \cap V) \xrightarrow{\delta} H_{dR}^p(S^n) \longrightarrow H_{dR}^p(U) \oplus H_{dR}^p(V)$$

Note that, since  $S^n$ 's are the one-point compactification of  $\mathbb{R}^n$ , we have that  $U \cong \mathbb{R}^n \cong V$ , and therefore,  $H_{dR}^p(U) = H_{dR}^{p-1}(U) = 0$  and  $H_{dR}^p(V) = H_{dR}^{p-1}(V) = 0$ . Also,  $U \cap V$  is just  $S^n \setminus \{N, S\}$  which is homeomorphic to  $\mathbb{R}^n \setminus \{0\}$ . Again,  $S^{n-1}$  is homotopically equivalent to  $\mathbb{R}^n \setminus \{0\}$  via the inclusion map  $i : S^{n-1} \hookrightarrow \mathbb{R}^n \setminus \{0\}$ . The homotopy inverse map is given by  $\gamma : \mathbb{R}^n \setminus \{0\} \mapsto S^{n-1}$  sending  $x$  to  $\frac{x}{\|x\|}$ . Clearly,  $\gamma \circ i \simeq id_{S^{n-1}}$ , and,  $i \circ \gamma \simeq id_{\frac{x}{\|x\|}}$ , which is further homotopically equivalent to  $id_{\mathbb{R}^n \setminus \{0\}}$  via the homotopy map  $H(x, t) = xt + (1-t)\frac{x}{\|x\|}$ . Hence,  $H_{dR}^{p-1}(U \cap V) = H_{dR}^{p-1}(S^{n-1})$ . Thus, the above sequence reduces down to the following exact sequence:

$$0 \longrightarrow H_{dR}^{p-1}(S^{n-1}) \xrightarrow{\delta} H_{dR}^p(S^n) \longrightarrow 0$$

from which the isomorphism  $H_{dR}^{p-1}(S^{n-1}) \cong H_{dR}^p(S^n)$  follows and this proves our claim.  $\square$

**Remark 7.1.0.2.** For  $p = n$ , any smooth orientation form induces a non-zero cohomology class which spans, and hence forms a basis of  $H_{dR}^n(S^n)$ .

Based on the cohomology of spheres, we state and prove the following lemma which will help us in understanding the next application of Mayer-Vietoris Theorem, that is the cohomology of the punctured Euclidean spaces.

**Lemma 7.1.0.3.** *If  $\eta$  is a top-form on  $S^n$ , then  $\eta$  is exact if and only if  $\int_{S^n} \eta = 0$ .*

*Proof.* Let us suppose,  $\eta \in \Omega^n(S^n)$  is exact. We know, by Stoke's theorem, the integral of an exact form over a manifold with empty boundary is 0. Since,  $\partial(S^n) = \emptyset$ , it follows that  $\int_{S^n} \eta = 0$ .

Conversely, let us consider  $\eta \in \Omega^n(S^n)$  such that  $\int_{S^n} \eta = 0$ . We need to prove that  $\eta$  is exact. By remark 7.1.0.2, let us suppose  $H_{dR}^n(S^n) = \text{span} \{[\omega]\}$ , where  $\omega$  is a smooth orientation form on  $S^n$ . Since,  $\eta$  is a top-form, and let us suppose  $\eta$  is not exact, then we have,  $\eta = c[\omega]$ , where  $c \in \mathbb{R}^*$ , that is,  $[\eta] = [c\omega]$ , that is for some exact form  $\lambda$ , we have  $\eta = c\omega + \lambda$ . It follows that

$$\int_{S^n} \eta = \int_{S^n} c\omega + \int_{S^n} \lambda = c \int_{S^n} \omega \neq 0$$

which is a contradiction. Hence,  $\eta$  is exact.  $\square$

## 7.2 Cohomology of punctured Euclidean spaces

**Theorem 7.2.0.1.** *Let  $n \geq 2$  and  $x \in \mathbb{R}^n$ . Let,  $M = \mathbb{R}^n \setminus \{x\}$  be the punctured Euclidean space. Then, the de Rham cohomology groups of  $M$  are given by*

$$H_{dR}^p(M) \cong \begin{cases} \mathbb{R} & \text{if } p = 0 \text{ or } p = n - 1 \\ 0 & \text{if otherwise} \end{cases}$$

A closed  $(n-1)$  form  $\eta$  is exact on  $M$  if and only if  $\int_S \eta = 0$  for some (and hence every)  $(n-1)$  dimensional sphere  $S \subset M$  centred at  $x$ .

*Proof.* For  $n \geq 1$ , we have already seen that the inclusion map  $i : S^{n-1} \hookrightarrow M$  is a homotopy equivalence, and thus the pullback map  $i^* : H_{dR}^p(M) \longrightarrow H_{dR}^p(S^{n-1})$  is an isomorphism. Henceforth, the cohomology of the punctured Euclidean spaces follows from the cohomology of the  $n$ -spheres.

Now, a closed  $(n-1)$ -form  $\eta$  is exact on  $M$  if and only if the closed pullback form  $i^*\eta$  is exact on  $S^{n-1}$  (since,  $S^{n-1}$  is a smooth embedding in  $M$ ), that is by lemma 7.1.0.3, if and only if  $\int_{S^{n-1}} i^*\eta = 0$ , that is,  $\int_{S^{n-1}} \eta = 0$ .  $\square$

**Lemma 7.2.0.2.** *The above theorem remains true if  $\mathbb{R}^n \setminus \{x\}$  is replaced by  $\mathbb{R}^n \setminus \bar{B}$  for some closed ball  $\bar{B} \subseteq \mathbb{R}^n$ .*

*Proof.* Let us consider a  $(n-1)$ -sphere  $S$  contained in  $\mathbb{R}^n \setminus \bar{B}$  centered at the origin. Since, this sphere is homeomorphic to  $S^{n-1}$  and the de Rham cohomology groups of homeomorphic spaces are equal, we have that  $H_{dR}^p(S) = H_{dR}^p(S^{n-1})$ . Henceforth, theorem 7.2.0.1 remains true if we consider the manifold  $M$  to be  $\mathbb{R}^n \setminus \bar{B}$  in place of  $\mathbb{R}^n \setminus \{x\}$ .

The second part of the theorem holds in the same way, by considering such a sphere  $S$  as above and following the lines of the proof of theorem 7.2.0.1.  $\square$

The following corollary generalizes theorem 7.2.0.1 from the Euclidean space  $\mathbb{R}^n$  to any open subset  $U$  of  $\mathbb{R}^n$ .

**Corollary 7.2.0.3.** *Let  $n \geq 2$ ,  $U \subseteq \mathbb{R}^n$  be an open and  $x \in U$ . Then  $H_{dR}^{n-1}(U \setminus \{x\}) \neq 0$ .*

*Proof.* Since,  $U$  is an open set,  $\exists$  a  $(n-1)$ -dimensional sphere  $S$  contained in  $U$  centered at  $x$ , that is  $S \subseteq U \setminus \{x\}$ . Let  $i : S \hookrightarrow U \setminus \{x\}$  be the inclusion map and let  $r : U \setminus \{x\} \rightarrow S$  be the radial projection onto  $S$ . Then, clearly  $r$  and  $i$  are smooth maps and  $r \circ i = id_S$ . This implies  $(r \circ i)^* = (id_S)^*$ , that is,  $i^* \circ r^* = id_{H_{dR}^{n-1}(S)}$ . Hence,  $r^* : H_{dR}^{n-1}(S) \rightarrow H_{dR}^{n-1}(U \setminus \{x\})$  is an injective map and since,  $H_{dR}^{n-1}(S) \neq 0$ , we have that  $H_{dR}^{n-1}(U \setminus \{x\}) \neq 0$ .  $\square$

## 8 Applications

In this section, we give some important applications of previous theorems and corollaries. We begin by stating the theorem of topological invariance of dimension. In the basic course of differential geometry we have learnt that any two diffeomorphic manifolds have the same topological dimension. But, using cohomology theory we can now prove a much stronger result as the following theorem will show.

**Theorem 8.0.0.1. (Invariance of dimension).** *Any two topological manifolds cannot be homeomorphic unless they have the same topological dimension.*

*Proof.* Let  $M$  be a  $m$ -manifold which is homeomorphic to a  $n$ -manifold. Then it implies that  $M$  itself is both an  $m$ -manifold and an  $n$ -manifold. We show that  $m = n$ . When either of  $m$  or  $n$  is 0, then  $M$  is just a discrete set of points, and thus the only neighbourhood of each point  $p \in M$  which is homeomorphic to  $\mathbb{R}^0$  is  $\{p\}$  itself and there is exactly one coordinate map  $\phi : \{p\} \rightarrow \mathbb{R}^0$ , and this implies  $M$  has a unique smooth structure. So, if  $m$  is 0, it is clear that  $n$  has to be 0. Now, without loss of generality, let us consider  $m > n \geq 1$ . Since,  $M$  is a  $m$ -manifold,  $\exists$  open subset  $V$  of  $M$  that is homeomorphic to  $\mathbb{R}^m$ . But again,  $M$  is a  $n$ -manifold, and  $V$  being open in  $M$ ,  $V$  itself is a  $n$ -manifold. Hence,  $\forall x \in V$ ,  $\exists$  an open neighbourhood  $U_x \subseteq V$  of  $x$  such that  $U_x \cong \mathbb{R}^n$ . We define a smooth structure on  $U_x$  via this homeomorphism, and since  $m-1 > n-1$ , by theorem 7.2.0.1 it follows that  $H_{dR}^{m-1}(U_x \setminus \{x\}) = 0$ . Again,  $U_x$  being an open subset of  $V$ , we have that  $U_x$  is homeomorphic to an open subset of  $\mathbb{R}^m$ . We give another smooth structure to  $U_x$  via this new homeomorphism, and then by corollary 7.2.0.3, it follows that  $H_{dR}^{m-1}(U_x \setminus \{x\}) \neq 0$ , which leads to a contradiction.  $\square$

Now, we prove a lemma which is an application of theorem 7.2.0.1, and will be used to calculate top degree cohomology groups. By Poincaré lemma, we know that any closed form defined on a star shaped domain is exact. The following lemma is much stronger version of it for  $\mathbb{R}^n$ , as it states, under a given condition, a compactly supported  $p$ -form on  $\mathbb{R}^n$  is the exterior derivative of a compactly supported  $(p-1)$ -form.

**Lemma 8.0.0.2. (Poincaré lemma with compact support).** *Let  $n \geq p \geq 1$ , and  $\omega$  is a compactly supported  $p$ -form on  $\mathbb{R}^n$ . If in addition,  $\int_{\mathbb{R}^n} \omega = 0$  for the case  $p = n$ , then there exists a compactly supported smooth  $(p-1)$ -form  $\eta$  on  $\mathbb{R}^n$  such that  $\omega = d\eta$ .*

*Proof.* **Case I:**  $n = p = 1$ .

In this case,  $\omega = f dx$ , for some compactly supported function  $f \in C^\infty(\mathbb{R})$ . We choose  $R > 0$  such that  $\text{supp } f \subseteq [-R, R]$ . Now, we define the following real valued function:

$$F : \mathbb{R} \rightarrow \mathbb{R}$$

$$x \mapsto \int_{-\infty}^x f(t) dt.$$

By fundamental theorem of calculus, it is easy to see that  $dF = F'(x) dx = f(x) dx = \omega$ . Also, it is straightforward to see that  $F$  is compactly supported, for if  $x < -R$ , then  $f(x) = 0$ , which clearly implies that  $F(x) = 0$ ,



and if  $x > R$ , then  $F(x) = \int_{-\infty}^x f(t)dt = \int_{-\infty}^{\infty} f(t) dt = 0$ , since,  $\int_{\mathbb{R}} f(t)dt = 0$ . Thus, in this case for a compactly supported 1-form  $\omega$ , we obtain a smooth function  $F$ , where  $\omega$  is the exterior derivative of  $F$  and  $\text{supp } F \subseteq [-R, R]$ .

**Case II:**  $n \geq 2$ .

Let  $B, B' \subseteq \mathbb{R}^n$  be open balls centered at the origin such that  $\text{supp } \omega \subseteq B \subseteq \bar{B} \subseteq B'$ . By our usual Poincaré lemma,  $\exists$  a smooth  $(p-1)$ -form  $\eta_0$  on  $\mathbb{R}^n$  such that  $\omega = d\eta_0$ . By using this, we will show that  $\omega$  is the exterior derivative of a compactly supported smooth form, by considering the following three subcases:

**Subcase I:**  $p = 1$ .

Since, support of  $\omega$  is contained within  $\bar{B}$ , we have that  $d\eta_0 = 0$  in  $\mathbb{R}^n \setminus \bar{B}$ . Again since,  $\mathbb{R}^n \setminus \bar{B}$  is connected for  $n \geq 2$ , it follows that  $\eta_0 = c$  in  $\mathbb{R}^n \setminus \bar{B}$ , for some constant  $c \in \mathbb{R}$ . Considering,  $\eta = \eta_0 - c$ , we have,  $d\eta = d(\eta_0 - c) = d\eta_0 = \omega$ . Thus, we obtain a compactly supported smooth form  $\eta$  such that  $\omega = d\eta$ .

**Subcase II:**  $1 < p < n$ .

$\eta_0|_{\mathbb{R}^n \setminus \bar{B}}$  is a closed  $(p-1)$  form. By theorem 7.2.0.1 and lemma 7.2.0.2, we have that  $H_{dR}^{p-1}(\mathbb{R}^n \setminus \bar{B}) = 0$ , since,  $p-1 < n-1$ . This implies that all  $(p-1)$  forms on  $\mathbb{R}^n \setminus \bar{B}$  are exact. Hence,  $\exists$  smooth  $(p-2)$  form  $\gamma$  such that  $\eta_0 = d\gamma$ . Now, let  $\psi$  be a smooth bump function supported in  $\mathbb{R}^n \setminus \bar{B}$  such that  $\psi = 1$  in  $\mathbb{R}^n \setminus B'$ . Consider  $\eta = \eta_0 - d(\psi\gamma)$ , which is smooth on all of  $\mathbb{R}^n$ . Then,  $\eta$  is compactly supported, for  $d(\psi\gamma) = d\gamma = \eta_0$  in  $\mathbb{R}^n \setminus B'$ , that is,  $\eta = 0$  in  $\mathbb{R}^n \setminus B'$  and also  $d\eta = d\eta_0 - d(d(\psi\gamma)) = \omega$ .

**Subcase III:**  $p = n$ .

In this case, we cannot use the same idea as that of subcase II, since  $H_{dR}^{n-1}(\mathbb{R}^n \setminus \bar{B}) \neq 0$ . From theorem 7.2.0.1, we know a closed form on  $\mathbb{R}^n \setminus \bar{B}$  is exact if its integral over any  $(n-1)$ -sphere centered at the origin and contained in  $\mathbb{R}^n \setminus \bar{B}$  is 0. Since,  $\omega$  is compactly supported, by Stoke's theorem, we have

$$0 = \int_{\mathbb{R}^n} \omega = \int_{B'} \omega = \int_{B'} d\eta_0 = \int_{\partial B'} \eta_0.$$

Since,  $\partial B'$  is a sphere centered at the origin and contained in  $\mathbb{R}^n \setminus B'$ , we have that  $\eta_0$  is exact in  $\mathbb{R}^n \setminus \bar{B}$ .  $\square$

## 9 de Rham cohomology of compactly supported forms

As the title suggests, in this section we are to define de Rham cohomology using compactly supported forms. If our manifold is compact, then this just reduces down to our usual notion of de Rham cohomology. But, in non-compact cases, the de Rham cohomology groups of the compactly supported forms are usually different from the ordinary cohomology groups. We are concerned about defining this generalization of cohomology groups, as in some cases, these groups are more useful than the usual de Rham cohomology groups.

**Definition 9.0.0.1.** Let  $M$  be a smooth manifold with or without boundary and let  $\Omega_c^p(M)$  denote the space of all compactly supported  $p$ -forms on  $M$ . The  **$p$ -th compactly supported de Rham cohomology group of  $M$**  is the quotient space

$$H_c^p(M) = \frac{\text{Ker}(d : \Omega_c^p(M) \rightarrow \Omega_c^{p+1}(M))}{\text{Im}(d : \Omega_c^{p-1}(M) \rightarrow \Omega_c^p(M))}.$$

For any topological spaces  $X$  and  $Y$ , a map  $f$  is said to be **proper** if  $f^{-1}(K)$  is compact for every compact set  $K \subseteq Y$ .

In our discussion of compactly supported de Rham cohomology groups, the notion of proper maps are very important because smooth maps do not pull back compactly supported forms to compactly supported ones, but *proper smooth* maps do. Thus, for a proper smooth map  $F : M \rightarrow N$ , there is an induced cohomology map  $F^* : H_c^p(N) \rightarrow H_c^p(M)$  for each  $p$ .



These compactly supported cohomology groups have a lot of applications; one might have seen them appearing in Poincaré duality in their algebraic topology course and furthermore, in geometry itself they are used to calculate the top degree cohomology groups. Now before we jump in our quest of classifying the top cohomology groups, we introduce a basic, but important notion of integration map, as follows:

### 9.1 The integration map

Let  $M$  be an oriented smooth  $n$ -manifold. Then there is a natural linear map that we can define:

$$\begin{aligned} I : \Omega_c^n(M) &\longrightarrow \mathbb{R} \\ \omega &\longmapsto \int_M \omega \end{aligned}$$

This map is linear by definition. Now, let us suppose  $\omega \in \mathcal{B}_c^n(M)$ . Then there exists  $\eta \in \Omega_c^{n-1}(M)$  such that  $\omega = d\eta$ . Since  $\eta$  is compactly supported, we can choose a compact subset  $K$  of  $M$  having a smooth boundary such that  $\text{supp } \eta \subseteq K$ . By Stoke's theorem,

$$\int_M \omega = \int_M d\eta = \int_K d\eta = \int_{\partial K} \eta = 0.$$

Hence, this  $I$  descends down to a linear map between  $H_c^n(M)$  and  $\mathbb{R}$ , denoted by the same symbol, which takes a cohomology class  $[\omega] \in H_c^n(M)$  to  $\int_M \omega$ .

**Example 9.1.0.1.**  $H_c^1(\mathbb{R}) \cong \mathbb{R}$ .

We know  $\mathbb{R}$  is an oriented 1-dimensional manifold. Hence, we have the following integration map:

$$\begin{aligned} I : H_c^1(\mathbb{R}) &\longrightarrow \mathbb{R} \\ [f(t)dt] &\longmapsto \int_{\mathbb{R}} f(t)dt. \end{aligned}$$

$I$  is surjective, for, any real  $c \in \mathbb{R}$ , there exists  $f \in C^\infty(\mathbb{R})$  such that  $\int_{\mathbb{R}} f(t)dt = c$ . We show that  $I$  is injective too.

$$\text{Ker } I = \{[f(t)dt] \in H_c^1(\mathbb{R}) : \int_{\mathbb{R}} f(t)dt = 0\}.$$

We define  $g(t) = \int_{-\infty}^t f(s)ds$ . Since  $f$  is compactly supported and  $\int_{\mathbb{R}} f(t)dt = 0$ , we have  $dg = f(t)dt$  and that  $g$  is compactly supported. Thus,  $f(t)dt \in \mathcal{B}_c^1(\mathbb{R})$  which implies  $[f(t)dt] = 0$  in  $H_c^1(\mathbb{R})$ . It follows that in this case,  $I$  is an isomorphism and hence, the result.

It is not just in the case of  $\mathbb{R}$ , but the top cohomology group of any compact manifold or the top degree compactly supported cohomology group of any manifold is isomorphic to  $\mathbb{R}$ .

## 10 Classification of top cohomologies of connected orientable manifolds

To classify the top cohomology groups of connected orientable manifolds, we divide into three cases, namely, compactly supported, compact and non-compact de Rham cohomology groups.

**Theorem 10.0.0.1. (Compactly supported case).** *If  $M$  is a connected oriented smooth  $n$ -manifold, then the integration map  $I : H_c^n(M) \rightarrow \mathbb{R}$  is an isomorphism, hence  $H_c^n(M)$  is 1-dimensional.*

*Proof.* **Case I:**  $n = 0$ .

Since, any 0-dimensional connected manifold is just a single point, say  $\{p\}$ , by corollary 3.2.0.2 it follows that  $H_c^0(\{p\})$  is the set of all compactly supported smooth functions from  $\{p\}$  to  $\mathbb{R}$ , that is, the set of all constant functions, thus,  $H_c^0(\{p\})$  is isomorphic to  $\mathbb{R}$ , and hence, 1-dimensional.

**Case II:**  $n \geq 1$ .

We need to prove the surjectivity and injectivity of the map  $I$ .

Given any real no.  $c$ , there exists a coordinate chart  $(U, (x^i))$  and a smooth bump function  $f$  such that the  $n$ -form  $\theta_0$  defined as

$$\theta_0 = \begin{cases} f dx^1 \wedge dx^2 \wedge \cdots \wedge dx^n & \text{if in } U \\ 0 & \text{if in } U^c \end{cases}$$

satisfies  $I(\theta_0) = \int_M f dx^1 \wedge dx^2 \wedge \cdots \wedge dx^n = c$ . Thus,  $I$  is surjective.

Now, to show that  $I$  is injective, we consider the kernel,  $\text{Ker } I = \{[\omega] \in H_c^n(M) \mid \int_M \omega = 0\}$ . It is clear that we need to show that if integral of any compactly supported orientation form  $\omega$  over  $M$  is 0, then,  $\exists \eta \in H_c^{n-1}(M)$  such that  $\omega = d\eta$ .

Since,  $M$  is a  $n$ -manifold, we consider a countable cover  $(U_i)_{i \in I}$  of  $M$ , where each  $U_i$  is an open subset of  $M$  diffeomorphic to  $\mathbb{R}^n$ . For each  $k$ , we define  $M_k = \cup_{i=1}^k U_i$ . By connectedness of  $M$ , we always have (by renumbering, if necessary)  $M_k \cap U_{k+1} \neq \emptyset$ . Now, every compactly supported  $n$ -form is supported in some  $M_k$ , for some finite  $k$  and let  $\omega \in \Omega_c^n(M)$  be arbitrary.

We claim that  $\int_{M_k} \omega = 0$  implies  $\omega = d\eta$  for some  $\eta \in \Omega_c^{n-1}(M_k)$ . We will prove this claim by induction, and it suffices to prove this claim because we can always extend  $\eta$  to the entire manifold by defining it to be 0 on  $M_k^c$ .

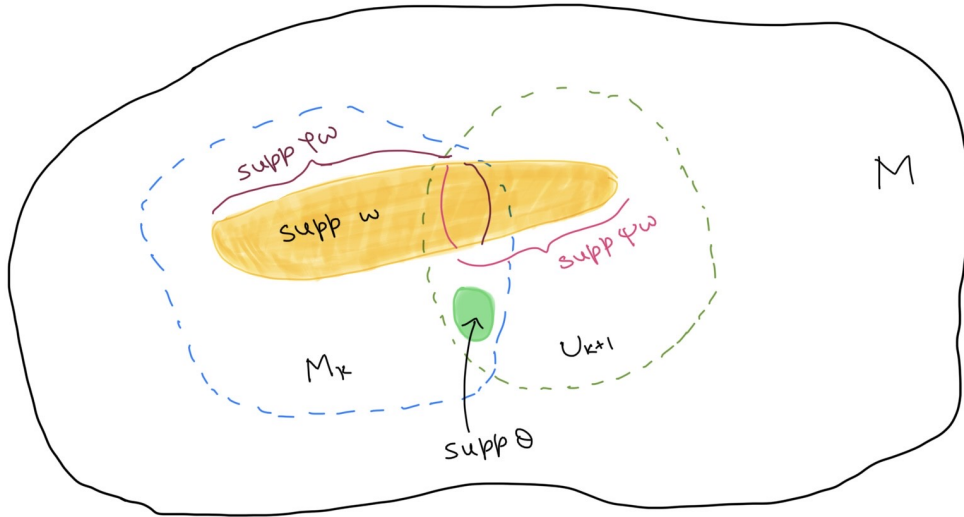


Figure 1: Computation of compactly supported top degree cohomology.

For  $k = 1$ , we have,  $M_1 = U_1$  which is diffeomorphic to  $\mathbb{R}^n$ , hence by lemma 8.0.0.2, our claim for this case holds. Now, let us assume the claim is true for  $k \geq 1$ . Let us suppose  $\omega$  is a compactly supported smooth  $n$ -form on  $M_{k+1}$  that satisfies  $\int_{M_{k+1}} \omega = 0$ , and also let us consider an auxiliary form  $\theta \in \Omega_c^n(M_{k+1})$  that is supported in  $M_k \cap U_{k+1}$  and it satisfies  $\int_{M_{k+1}} \theta = 1$ . (Such a form is easily constructed by considering bump functions in terms of above coordinates.) Let  $\{\phi, \psi\}$  be a partition of unity subordinate to the open cover  $\{M_k, U_{k+1}\}$  and suppose

$c = \int_{M_{k+1}} \phi\omega$ . Then, it implies  $c \int_{M_{k+1}} \theta = \int_{M_{k+1}} \phi\omega$ , that is,  $\int_{M_{k+1}} (\phi\omega - c\theta) = 0$ . Now, since a manifold is locally compact, the support of any partition of unity defined on the manifold subordinate to an open cover is always compact. Hence,  $\text{supp } \phi\omega$  and  $\text{supp } \psi\theta$  are compact and, it is a clear observation that

$$\begin{aligned} \text{supp } (\phi\omega - c\theta) &\subseteq (\text{supp } \phi\omega) \cup (\text{supp } c\theta) \\ \text{supp } (\psi\omega + c\theta) &\subseteq (\text{supp } \psi\omega) \cup (\text{supp } c\theta), \end{aligned}$$

and since, supports are always closed, being closed subsets of compact sets, we have that  $\text{supp } (\phi\omega - c\theta)$  and  $\text{supp } (\psi\omega + c\theta)$  are compactly supported  $n$ -forms on  $M_k$  and  $U_{k+1}$  respectively. So, by induction hypothesis,  $\int_{M_{k+1}} (\phi\omega - c\theta) = 0$ , that is,  $\int_{M_k} (\phi\omega - c\theta) = 0$  implies that there exists  $\eta \in \Omega_c^{n-1}(M_k)$  such that  $d\eta = \phi\omega - c\theta$ .

Also, we have that

$$\begin{aligned} \int_{M_{k+1}} (\psi\omega + c\theta) &= \int_{M_{k+1}} (1 - \phi)\omega + c \int_{M_{k+1}} \theta \\ &= \int_{M_{k+1}} \omega - \int_{M_{k+1}} \phi\omega + \int_{M_{k+1}} c\theta \\ &= 0, \end{aligned}$$

that is, we have  $\int_{U_{k+1}} (\psi\omega + c\theta) = 0$ . Hence, by lemma 8.0.0.2, we have that there exists  $\zeta \in \Omega_c^{n-1}(U_{k+1})$  such that  $d\zeta = \psi\omega + c\theta$ . Now, clearly,  $(\eta + \zeta) \in \Omega_c^{n-1}(M_{k+1})$  and  $d(\eta + \zeta) = (\phi\omega - c\theta + \psi\omega + c\theta) = \phi\omega + \psi\omega = (\phi + \psi)\omega = \omega$ , and this completes our inductive step. Thus,  $I$  is injective.  $\square$

**Corollary 10.0.0.2. (Compact case).** *If  $M$  is a compact, connected, orientable smooth  $n$ -manifold, then  $H_{dR}^n(M)$  is 1-dimensional, and is spanned by the cohomology class of any smooth orientation form.*

*Proof.* Since  $M$  is a compact manifold, from the previous theorem it follows that  $H_{dR}^n(M) = H_c^n(M)$ , and the integral of any smooth orientation form over the manifold  $M$  is non-zero.  $\square$

**Theorem 10.0.0.3. (Compactly supported cohomology of  $\mathbb{R}^n$ ).** *For  $n \geq 1$ , the compactly supported de Rham cohomology groups of  $\mathbb{R}^n$  are*

$$H_c^p(\mathbb{R}^n) \cong \begin{cases} 0 & \text{if } 0 \leq p \leq n-1, \\ \mathbb{R} & \text{if } p = n. \end{cases}$$

*Proof.* Let  $p = 0$ . Then,

$$H_c^0(\mathbb{R}^n) = \frac{\mathcal{Z}_c^0(\mathbb{R}^n)}{\mathcal{B}_c^0(\mathbb{R}^n)} = \{f \in C_c^\infty(\mathbb{R}^n) : df = 0\},$$

where  $C_c^\infty(\mathbb{R}^n)$  denotes the set of all compactly supported smooth functions on  $\mathbb{R}^n$ . Now,  $df = 0$  on  $\mathbb{R}^n$ , implies that  $f \equiv c$ , for some constant  $c \in \mathbb{R}$ . Suppose,  $c \neq 0$ , then  $f$  takes a non-zero constant value on entire  $\mathbb{R}^n$ , but since  $\mathbb{R}^n$  is not compact, it implies that  $f$  is not compactly supported, which leads to a contradiction. Hence,  $f \equiv 0$ . Since,  $f \in C_c^\infty(\mathbb{R}^n)$  was arbitrary, it follows that  $H_c^0(\mathbb{R}^n) = 0$ .

Now, for  $1 \leq p \leq n-1$ , if  $\omega \in \mathcal{Z}_c^p(\mathbb{R}^n)$ , then by lemma 8.0.0.2, we have that  $\exists \eta \in \Omega_c^{p-1}(\mathbb{R}^n)$  such that  $\omega = d\eta$ . Hence, in this case,  $\mathcal{Z}_c^p(\mathbb{R}^n) = \mathcal{B}_c^p(\mathbb{R}^n)$ , which implies that  $H_c^p(\mathbb{R}^n) = 0$ .

The case  $p = n$  is already covered in theorem 10.0.0.1.  $\square$

Next, before we classify the top cohomology of orientable non-compact case, we define and prove the existence of smooth exhaustion function on a smooth manifold.

**Definition 10.0.0.4. (Exhaustion function).** For a smooth manifold  $M$ , a function  $f : M \rightarrow \mathbb{R}$  is an exhaustion function for  $M$  if the set  $f^{-1}((-\infty, c])$  is compact for each  $c \in \mathbb{R}$ .

**Theorem 10.0.0.5. (Existence of smooth exhaustion function).** *Every smooth manifold with or without boundary admits a smooth positive exhaustion function.*

*Proof.* Every smooth manifold admits a countable open cover by pre-compact open sets. Let  $M$  be a smooth manifold and let  $\{V_j\}_{j=1}^\infty$  be such an open cover for  $M$ . Let  $\{\psi_j\}$  be a partition of unity subordinate to this open cover.

We define

$$\begin{aligned} f : M &\longrightarrow \mathbb{R} \\ p &\longmapsto \sum_{j=1}^{\infty} j\psi_j(p) \end{aligned}$$

It is clear that in a neighbourhood of any point on the manifold, only finitely many terms of the summation is non-zero, hence,  $f$  is smooth on  $M$ . Also,  $f$  is positive because  $f(p) \geq \sum_{j=1}^{\infty} \psi_j(p) = 1$ . Now, we are just left to show that  $f$  is an exhaustion function.

Let  $c \in \mathbb{R}$  be arbitrary. Let us choose a positive integer  $N > c$ . For any point  $p \in M$ , we claim that  $f(p) \leq c$  implies that  $p \in \cup_{j=1}^N \bar{V}_j$ , proving which would suffice, as it would imply  $f^{-1}((-\infty, c])$  is a closed subset of the compact set  $\cup_{j=1}^N \bar{V}_j$ , and is therefore compact. We prove this claim in a contrapositive way. Let us suppose  $p \notin \cup_{j=1}^N \bar{V}_j$ , then,  $\psi_j(p) = 0$  for all  $j = 1, 2, \dots, N$ . Then,

$$f(p) = \sum_{j=N+1}^{\infty} j\psi_j(p) \geq N \sum_{j=N+1}^{\infty} \psi_j(p) = N \sum_{j=1}^{\infty} \psi_j(p) = N > c.$$

This proves the claim.  $\square$

**Theorem 10.0.0.6. (Non-compact case).** *If  $M$  is a non-compact, orientable, smooth  $n$ -manifold, then,  $H_{dR}^n(M) = 0$ .*

*Proof.* Let us choose an orientation on  $M$ . Let  $f \in C^\infty(M)$  be a smooth positive exhaustion function on  $M$ , whose existence is ensured by the above theorem. Since,  $f$  is positive,  $\inf_M f$  exists and subtracting by a suitable constant, we can have  $\inf_M f = 0$ . The connectedness and the non-compactness of the manifold  $M$  imply that  $f(M) = [0, \infty)$ .

Let us now define  $V_i = f^{-1}((i-2, i))$ . It is easy to see that  $\bar{V}_i$  is compact, for each  $i$ , using the fact that  $f^{-1}$  is an exhaustion function. By definition of  $V_i$ , it is clear that  $\{V_i\}_{i=1}^\infty$  forms an open cover of  $M$  by pre-compact open sets, where  $V_i \cap V_j \neq \emptyset$  iff  $j = i-1, i, i+1$ . Let  $\{\psi_i\}$  be a smooth partition of unity subordinate to the open cover. Now, for each  $i$ , let us consider the auxiliary forms  $\theta_i \in \Omega_c^n(M)$  such that  $\text{supp } \theta_i \subseteq V_i \cap V_{i+1}$  and  $\int_M \theta_i = 1$ . Suppose  $\omega$  is a compactly supported smooth  $n$ -form on  $M$ . We define  $\omega_i = \psi_i \omega$ , thus,  $\omega_i \in \Omega_c^n(V_i)$ . Let  $c_1 = \int_{V_1} \omega_1$ . As shown in the previous theorem, it is clear that  $\omega_1 - c_1 \theta_1 \in \Omega_c^n(V_1)$  and  $\int_M (\omega_1 - c_1 \theta_1) = 0$  and hence,  $\exists \eta_1 \in \Omega_c^{n-1}(V_1)$  such that  $d\eta_1 = \omega_1 - c_1 \theta_1$ . Next we choose  $c_2 \in \mathbb{R}$  such that  $\int_{V_2} (\omega_2 + c_1 \theta_1 - c_2 \theta_2) = 0$  and by similar arguments,  $\exists \eta_2 \in \Omega_c^n(V_2)$  such that  $d\eta_2 = \omega_2 + c_1 \theta_1 - c_2 \theta_2$ . Proceeding by induction,  $\exists \eta_j \in \mathbb{R}$  such that  $d\eta_j = \omega_j + c_{j-1} \theta_{j-1} - c_j \theta_j$ . Expanding  $\eta_j$  to the entire manifold  $M$  by defining it to be 0 on  $M \setminus V_j$ , we set  $\eta = \sum_{j=1}^\infty \eta_j$ . By definition of  $\eta$ , it is clear that  $\eta$  is smooth on  $M$ , since on each  $V_j$ , at most three  $\eta_j$ 's are non-zero, and thus, on taking the exterior derivative of  $\eta$ , it is a direct observation that  $d\eta = \sum_{j=1}^\infty \omega_j = \omega$ . Henceforth, it follows that every smooth  $n$ -form on  $M$  is exact.  $\square$

## 11 Classification of top cohomology of connected non-orientable manifolds

Now, before we move on to prove the top cohomology for non-orientable manifolds, we present a lemma which will be the key ingredient in proving the aforementioned.

**Lemma 11.0.0.1.** *Suppose  $M$  is a smooth non-orientable connected manifold and  $\hat{\pi} : \hat{M} \rightarrow M$  be its orientation covering. For each  $p$ , the induced cohomology maps  $\hat{\pi}^* : H_{dR}^p(M) \rightarrow H_{dR}^p(\hat{M})$  and  $\hat{\pi} : H_c^p(M) \rightarrow H_c^p(\hat{M})$  are injective.*

*Note:* Due to lemma A.3.0.3, the map between the compactly supported de Rham cohomologies is well defined in this present lemma.

*Proof.* First, we prove the theorem for the compactly supported case. Let  $\omega$  be a closed comapactly supported  $p$ -form on  $M$  such that  $\pi^*[\omega] = 0$  in  $H_c^p(\hat{M})$ , that is, there exists  $\eta \in \Omega_c^{p-1}(\hat{M})$  such that  $d\eta = \hat{\pi}^*\omega$ .

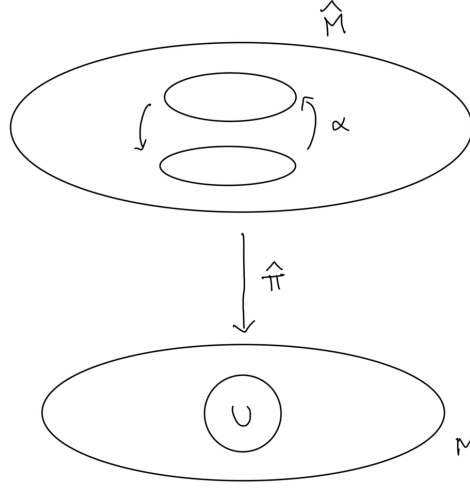


Figure 2: Automorphisms of the orientation covering.

Since, the orientation covering is unique upto an isomorphism, let us consider the unique non-trivial covering automorphism of  $\hat{M}$ ,  $\alpha : \hat{M} \rightarrow \hat{M}$  and its induced cohomology map  $\alpha^* : H_{dR}^p(\hat{M}) \rightarrow H_{dR}^p(\hat{M})$ .

Define  $\tilde{\eta} = \frac{1}{2}(\eta + \alpha^*\eta)$ , which is clearly compactly supported. Since,  $\hat{\pi} \circ \alpha = \hat{\pi}$ , we have that

$$d\tilde{\eta} = \frac{1}{2}(d\eta + d\alpha^*\eta) = \frac{1}{2}(\hat{\pi}^*\omega + \alpha^*\hat{\pi}^*\omega) = \hat{\pi}^*\omega.$$

Let  $U \subseteq M$  be an evenly covered open subset of  $M$ . By proposition A.3.0.1, since  $\hat{M}$  is a two-sheeted covering space of  $M$ , any two local sections  $\sigma_1$  and  $\sigma_2$  over  $U$  are related by  $\sigma_2 = \alpha \circ \sigma_1$ . Note that

$$\sigma_2^*\tilde{\eta} = (\alpha \circ \sigma_1)^*\tilde{\eta} = \sigma_1^*\alpha^*\tilde{\eta} = \sigma_1^*\tilde{\eta},$$

as  $\alpha \circ \alpha = \text{id}_{\hat{M}}$ , and

$$\alpha^*\tilde{\eta} = \frac{1}{2}(\alpha^*\eta + (\alpha \circ \alpha)^*\eta) = \frac{1}{2}(\alpha^*\eta + \eta) = \tilde{\eta}.$$

Hence, for any smooth local section  $\sigma : U \rightarrow \hat{M}$ , we can define a global  $(p-1)$ -form  $\beta$  on  $M$  such that  $\beta|_U = \sigma^*\tilde{\eta}$ .

Note that here we constructed the form  $\tilde{\eta}$  just so that we could ensure various definitions of pullback of local sections agree on the overlap of evenly covered open sets. If we would have just worked with  $\eta$ , then there is no guarantee that  $\sigma_1^*\eta$  would be same as  $\sigma_2^*\eta$  as  $\alpha^*\eta$  is not necessarily  $\eta$ .

Now, since,  $\text{supp } \beta = \hat{\pi}(\text{supp } \tilde{\eta})$ ,  $\beta$  is compactly supported. To compute the exterior derivative of  $\beta$ , given any  $p \in M$ , choose a evenly covered open set  $U_p$  containing  $p$  and a local section  $\sigma : U_p \rightarrow \hat{M}$ . Then, we have,

$$d\beta = d\sigma^*\tilde{\eta} = \sigma^*\hat{\pi}^*\omega = (\hat{\pi} \circ \sigma)^*\omega = \omega,$$

as  $\sigma$  being a local section, we have that  $\hat{\pi} \circ \sigma = \text{id}_U$ . Thus,  $\omega$  is exact, which implies that  $[\omega] = 0$  in  $H_c^p(M)$ . therefore, the map  $\hat{\pi} : H_c^p(M) \rightarrow H_c^p(\hat{M})$  is injective.

The same arguments hold in the case of ordinary de Rham cohomology with all references to compactly supported deleted.  $\square$

**Theorem 11.0.0.2. (Non-orientable case)** *Let  $M$  be a connected, non-orientable smooth manifold, then,  $H_c^n(M) = 0$  and  $H_{dR}^n(M) = 0$ .*

*Proof.* Let us first consider the case of compactly supported cohomology. By the preceeding lemma, it is enough to show that the map  $\hat{\pi}^* : H_c^n(M) \rightarrow H_c^n(\hat{M})$  is the zero map, where  $\hat{\pi} : \hat{M} \rightarrow M$  is the orientation covering of  $M$ . As in the preceeding proof, we consider the non-trivial covering automorphism  $\alpha : \hat{M} \rightarrow \hat{M}$ . By theorem A.3.0.5, it follows that the action of  $\text{Aut}_{\hat{\pi}} \hat{M}$  on  $\hat{M}$  cannot be orientation preserving. Since,  $\hat{M}$  is a connected two-sheeted covering space of  $M$ ,  $\text{Aut}_{\hat{\pi}} \hat{M}$  is just the two element group  $\{\text{Id}_{\hat{M}}, \alpha\}$ , and thus,  $\alpha$  cannot be orientation preserving. So, we have that  $\alpha$  is orientation reversing. Suppose  $\omega$  is any compactly supported  $n$ -form and let  $\hat{\pi}^*\omega$ . By lemma A.3.0.3, since,  $\hat{\pi}$  is a proper map,  $\hat{\omega}$  is compactly supported and  $\hat{\pi} \circ \alpha = \hat{\pi}$  implies that

$$\alpha^*\hat{\omega} = \alpha^*(\hat{\pi}^*\omega) = (\hat{\pi} \circ \alpha)^*\omega = \hat{\pi}^*\omega = \hat{\omega}.$$

Now, because  $\alpha$  is orientation reversing, we have that

$$\int_{\hat{M}} \hat{\omega} = - \int_{\hat{M}} \alpha^*\hat{\omega} = - \int_{\hat{M}} \hat{\omega},$$

that is,  $\int_{\hat{M}} \hat{\omega} = 0$ . Since,  $\hat{M}$  is a connected, orientable smooth manifold and  $\hat{\omega} \in H_c^n(\hat{M})$ , by theorem 10.0.0.1,  $\int_{\hat{M}} \hat{\omega} = 0$  implies that  $[\omega] = 0$  in  $H_c^n(M)$ . This proves that  $H_c^n(M) = 0$ .

For the non-compact case, that is when  $M$  is non-compact,  $\hat{M}$  is also non-compact, otherwise, being a continuous image of a compact set,  $M$  would become compact. By theorem 10.0.0.6, we already have  $H_{dR}^n(\hat{M}) = 0$ , and by injectivity of the map  $\hat{\pi}^* : H_{dR}^n(M) \rightarrow H_{dR}^n(\hat{M})$ , it follows that  $H_{dR}^n(M) = 0$ .  $\square$

## A Fundamentals of Differential Geometry

### A.1 Flows and Integral Curves

**Definition A.1.0.1. (Flow domain).** A flow domain is a subset  $\mathcal{D} \subset \mathbb{R} \times M$  such that for any  $p \in M$ , the set  $\mathcal{D}^{(p)} = \{t \in \mathbb{R} : (t, p) \in \mathcal{D}\}$  is an open set containing 0.

Let  $\theta : \mathcal{D} \rightarrow M$  be a flow.

**Definition A.1.0.2. (Infinitesimal generator).** For each  $p \in M$ , the assignment  $p \mapsto V_p$  is a rough vector field on  $M$ , where  $V_p = \theta^{(p)'}(0)$  is said to be the infinitesimal generator of  $\theta$ .

**Definition A.1.0.3. (Maximal integral curve).** An integral curve defined on an open interval, say  $\mathcal{D}^{(p)}$ , which cannot be extended to an integral curve defined on another open interval  $\mathcal{D}^{(p)'}$  such that  $\mathcal{D}^{(p)} \subset \mathcal{D}^{(p)'}$  is a maximal integral curve.

Quite intuitively, a **maximal flow** is a flow that cannot be extended to a flow with larger flow domain.

**Proposition A.1.0.4.** If  $\theta : \mathcal{D} \rightarrow M$  is a smooth flow, then the vector field  $V$  which is the infinitesimal generator of  $\theta$  is smooth, and for each  $p$  in  $M$ ,  $\theta^{(p)}$  is an integral curve of  $V$ .

**Theorem A.1.0.5. (Fundamental theorem of flows).** For every smooth vector field defined on a smooth manifold  $M$ , there is a unique smooth maximal flow  $\theta : \mathcal{D} \rightarrow M$  whose infinitesimal generator is  $V$ .

The above two results help in giving the following one-to-one correspondence:

$$\left\{ \text{smooth vector fields} \right\} \longleftrightarrow \left\{ \text{smooth flows} \right\}$$

### A.2 Lie derivatives

#### A.2.1 Lie derivatives of vector fields

Let  $M$  be a smooth manifold with or without boundary. Let  $V$  and  $W$  be two smooth vector fields on  $M$  with  $\theta$  being the flow of  $V$ . We denote  $\mathcal{L}_V W$  as the *lie derivative of  $W$  wrt  $V$*  and for any  $p \in M$ , we define it as

$$\begin{aligned} (\mathcal{L}_V W)_p &= \left. \frac{d}{dt} \right|_{t=0} d(\theta_{-t})_{\theta_t(p)} W_{\theta_t(p)} \\ &= \lim_{t \rightarrow 0} \frac{d(\theta_{-t})_{\theta_t(p)} W_{\theta_t(p)} - W_p}{t} \end{aligned}$$

provided that the derivative exists. If  $\partial M \neq \emptyset$ , then  $\mathcal{L}_V W$  makes sense as long as  $V$  is tangent to  $\partial M$ .

**Lemma A.2.1.1.** Suppose  $M$  is a smooth manifold with or without boundary, and  $V, W \in \mathfrak{X}(M)$ . If  $\partial M \neq \emptyset$ , assume in addition that  $V$  is tangent to  $\partial M$ . The  $(\mathcal{L}_V W)_p$  exists for every  $p \in M$  and is a smooth vector field.

#### A.2.2 Lie derivative of differential forms

Let  $M$  be a smooth manifold with or without boundary. Let  $V$  be a smooth vector field and  $A$  be a smooth covariant tensor field defined on  $M$ . If  $\partial M \neq \emptyset$ , then assume in addition that  $V$  is tangent to  $\partial M$ . If  $t$  is sufficiently close to 0, then  $\theta_t$  is a diffeomorphism from a neighbourhood of  $p$  to a neighbourhood of  $\theta_t(p)$ .

Here,  $d(\theta_t)_p^* A_{\theta_t(p)}$  is just  $(\theta_t^* A)_p$ , that is it pulls back tensors at  $\theta_t(p)$  to ones at  $p$  and is given by the formula,

$$d(\theta_t)_p^* A_{\theta_t(p)}(v_1, \dots, v_k) = A_{\theta_t(p)}(d(\theta_t)_p v_1, \dots, d(\theta_t)_p v_k).$$

For any  $p \in M$ , we define the *Lie derivative of  $A$  wrt  $V$*  as

$$\begin{aligned} (\mathcal{L}_V A)_p &= \left. \frac{d}{dt} \right|_{t=0} (\theta_t^* A)_p \\ &= \lim_{t \rightarrow 0} \frac{d(\theta_t)_p^* A_{\theta_t(p)} - A_p}{t} \end{aligned}$$

provided that the derivative exists.

**Remark A.2.2.1.** In this case of lie derivative of differential forms, there is a same analogue of lemma A.2.1.1.

**Proposition A.2.2.2.** *In the above setting, for any  $(t_0, p)$  in the domain of  $\theta$ , we have*

$$\left. \frac{d}{dt} \right|_{t=t_0} (\theta_t^* A)_p = (\theta_{t_0}^* (\mathcal{L}_V A))_p \quad (9)$$

*Proof.* By definition of pullback, let us expand (9) as follows:

$$\left. \frac{d}{dt} \right|_{t=t_0} d(\theta_t)^*(A_{\theta_t(p)}) = d(\theta_{t_0})^*(\mathcal{L}_V A)_{\theta_{t_0}(p)}.$$

To prove the above, we make a change of variable by setting  $t = s + t_0$ . Then,

$$\begin{aligned} \left. \frac{d}{dt} \right|_{t=t_0} d(\theta_t)^*(A_{\theta_t(p)}) &= \left. \frac{d}{ds} \right|_{s=0} d(\theta_{s+t_0})^*(A_{\theta_{s+t_0}(p)}) \\ &= \left. \frac{d}{ds} \right|_{s=0} d(\theta_{t_0})^* d(\theta_s)^*_{\theta_{t_0}(p)} (A_{\theta_s(\theta_{t_0}(p))}) \\ &= d(\theta_{t_0})^* \left. \frac{d}{ds} \right|_{s=0} d(\theta_s)^*_{\theta_{t_0}(p)} (A_{\theta_s(\theta_{t_0}(p))}) \\ &= d(\theta_{t_0})^* (\mathcal{L}_V A)_{\theta_{t_0}(p)} \\ &= (\theta_{t_0}^* (\mathcal{L}_V A))_p. \end{aligned}$$

□

### A.3 Further results on orientations

Before we state the main theorem of this section, that is, the *orientation covering theorem*, we give some definitions: Let  $N$  and  $M$  be two smooth manifolds with or without boundary, then  $\pi : N \rightarrow M$  is called a **generalised covering map**, if  $\pi$  satisfies all the properties of a covering map except that  $N$  might *not be connected*. Moreover,  $\pi$  is said to be a **generalised smooth covering map** if  $\pi$  is a local diffeomorphism, that is, for any  $p$  in  $M$  if  $U_p$ , an open set in  $M$  containing  $p$  is such that  $\pi^{-1}(U_p) = \bigsqcup_{i=1}^n V_i$ , where  $V_i$ 's are open subsets of  $N$ , then for each  $i$ ,  $\pi|_{V_i} : V_i \rightarrow U_p$  is a diffeomorphism.

Now, we define what is called an *orientation covering*. Let  $M$  be a connected, smooth, positive dimensional manifold. Let

$$\hat{M} = \{(p, \theta_p) : p \in M \text{ and } \theta_p \text{ is an orientation of } T_p M\}.$$

Define

$$\begin{aligned} \hat{\pi} : \hat{M} &\longrightarrow M \\ (p, \theta_p) &\longmapsto p. \end{aligned}$$

where for each  $p$ , the cardinality of the fibre  $\pi^{-1}(p)$  is 2, since, every vector space, in particular, every tangent space has just two orientations. This map  $\hat{\pi} : \hat{M} \rightarrow M$  is called the **orientation covering** of  $M$ .

**Proposition A.3.0.1. (Properties of the orientation covering).** *Suppose  $M$  is a connected, smooth, positive dimensional manifold with or without boundary, and, let  $\hat{\pi} : \hat{M} \rightarrow M$  be its orientation covering. Then  $\hat{M}$  can be given the structure of a smooth, oriented manifold with or without boundary, with the following properties:*

1.  $\hat{\pi} : \hat{M} \rightarrow M$  is a generalised smooth covering map.
2. A connected open subset  $U \subseteq M$  is evenly covered by  $\hat{\pi}$  iff  $U$  is orientable.
3. If  $U \subseteq M$  is an evenly covered open subset, then, every orientation on  $U$  is the pullback orientation induced by the local section of  $\hat{\pi}$  over  $U$ .

Now, we state the main result of this section:



**Theorem A.3.0.2. (Orientation covering theorem).** *Let  $M$  be a connected smooth manifold with or without boundary and let,  $\hat{\pi} : \hat{M} \rightarrow M$  be its orientation covering.*

1. *If  $M$  is orientable, then,  $\hat{M}$  has exactly two components, and the restriction of  $\hat{M}$  onto each of these components is a diffeomorphism onto  $M$ .*
2. *If  $M$  is non-orientable, then,  $\hat{M}$  is connected and  $\hat{\pi}$  is a two-sheeted smooth covering map.*

*Proof.* Proof of (1): By proposition A.3.0.1.2, we have that  $M$  is evenly covered by  $\hat{\pi}$ , that is,  $\pi^{-1}(M) = \hat{M}$  is the disjoint union of two of its open sets, which implies that  $\hat{M}$  has exactly two components. Also, since  $\hat{\pi}$  is a smooth covering map, the restriction of  $\hat{\pi}$  to each of the two disjoint components of  $\hat{M}$  is a diffeomorphism onto  $M$ .

Proof of (2): Let us suppose  $\hat{M}$  is not connected. Let  $W$  be a component of  $M$  and since our manifold  $M$  is connected, by a basic result in topology, we have that  $\hat{\pi}|_W$  is a covering map and all of its fibers have the same cardinality. Since, the fibers of  $\hat{\pi}$  have cardinality 2, it follows that for each  $p \in M$ ,  $\hat{\pi}|_W^{-1}(p)$  has either 1 or 2 as its cardinality. Suppose, cardinality is 1, then  $\hat{\pi}|_W$  would be a smooth injective covering map, hence a diffeomorphism. The restriction of  $\hat{\pi}$  to each of the connected components of  $\hat{\pi}$  would induce an orientation on  $M$ . But, since  $M$  is non-orientable, it follows that the cardinality of the fibers of  $\hat{\pi}|_W$  must be 2, and thus  $W = \hat{M}$ . So, we have that  $\hat{M}$  is connected. Since, every connected manifold has exactly two orientations,  $\hat{\pi}$  is a smooth two-sheeted covering map.  $\square$

Now, using the above theorem, we want to give a necessary and sufficient condition for a manifold to be orientable, but for that we also need another lemma, which is stated and proved as follows:

**Lemma A.3.0.3.** *A finite sheeted covering map is proper.*

*Proof.* Let  $p : \tilde{X} \rightarrow X$  be a finite sheeted covering map. Let  $K \subseteq Y$  be any compact set. To show that  $p$  is proper, we need to show that  $p^{-1}(K)$  is compact in  $\tilde{X}$ .

Let  $\{U_\alpha\}$  be an open cover of  $p^{-1}(K)$ . Since, covering maps are open maps,  $\{p(U_\alpha)\}$  is an open cover of  $K$ . Since,  $p$  is a covering map, for each  $x \in M$  there exists an evenly covered open set  $U_x$  and also, since  $p$  is finite sheeted, it is not hard to observe that  $p^{-1}(U_x)$  is a finite disjoint union of open sets in  $\hat{M}$ , that is say,  $p^{-1}(U_x) = \bigsqcup_{i=1}^{n_x} V_i^x$ . Now, for every such evenly covered open set  $U_x$ , whenever  $V_i^x$  intersect  $U_\alpha$ 's non-trivially, we can shrink  $V_i^x$  for each  $i$  so that it is contained in some  $U_\alpha$ . Clearly, we can do this shrinking process only because the disjoint union is finite. These  $\{U_x\}_{x \in K}$  form an open open cover of  $K$  and by compactness of  $K$ , we can cover it by finitely many of these, say,  $\{U_{x_j}\}_{j=1}^m$ . Hence,  $K \subseteq \cup_{j=1}^m U_{x_j}$ , which implies  $p^{-1}(K) \subseteq p^{-1}(\cup_{j=1}^m U_{x_j}) = \cup_{j=1}^m p^{-1}(U_{x_j})$ . By the above discussion it is clear that this finite union forms a finite subcollection of  $\{U_\alpha\}$ . Thus,  $p^{-1}(K)$  is compact.  $\square$

**Definition A.3.0.4. (Orientation preserving group action).** If  $G$  is a lie group acting smoothly on a smooth manifold  $E$ , we say that the action is orientation preserving if for each  $g \in G$ , the diffeomorphism  $x \mapsto g \cdot x$  is orientation preserving.

**Theorem A.3.0.5.** *Suppose  $E$  is a connected, oriented manifold with or without boundary and  $\pi : E \rightarrow M$  is a smooth normal covering map. Then  $M$  is orientable if and only if the action of  $\text{Aut}_\pi(E)$  on  $E$  is orientation preserving.*

*Proof.* Let us first suppose that  $M$  is orientable. Let  $\mathcal{O}_E$  be the orientation on  $E$ . We know that any connected manifold has exactly two orientations, so, of the two orientations on  $M$ , let  $\mathcal{O}_M$  be the orientation which makes the map  $d\pi_q : T_q E \rightarrow T_{\pi(q)} M$  orientation preserving. It is for the uniqueness of a pullback orientation, that given any orientation on  $E$ , we already have the orientation on  $M$  which makes the map  $d\pi_q$  orientation preserving. We know that any two orientations agreeing at a point are equal, and since,  $\pi^* \mathcal{O}_M$  agrees with  $\mathcal{O}_E$  at the point  $q$ , it follows that  $\pi^* \mathcal{O}_M = \mathcal{O}_E$ . Let  $\phi \in \text{Aut}_\pi(E)$ , and since

$$\phi^*(\mathcal{O}_E) = \phi^*(\pi^* \mathcal{O}_M) = (\pi \circ \phi)^* \mathcal{O}_M = \pi^* \mathcal{O}_M = \mathcal{O}_E,$$

we have that the action of  $\text{Aut}_\pi(E)$  on  $E$  is orientation preserving.

Conversely, let us suppose the action is orientation preserving and let  $p \in M$  be arbitrary. Let  $U_p$  be an evenly covered open subset containing  $p$  and let  $\sigma : U_p \rightarrow E$  be a local section of  $\pi$  which induces an orientation  $\sigma^* \mathcal{O}_E$  on  $U$ . Now, let  $\sigma_1 : U_p \rightarrow E$  be another local section over  $U$ . Since,  $\pi$  is a normal covering map,  $\text{Aut}_\pi(E)$  acts

transitively on each fiber of  $\pi$ , thus, there exists a covering automorphism  $\phi$  such that  $\sigma$  and  $\sigma_1$  are related by  $\sigma_1 = \phi \circ \sigma$  on the neighbourhood  $U_p$ . Hence,  $\phi \circ \sigma$  is a local section of  $\pi$  that agrees with  $\sigma_1$  on all of  $U_p$ . Since,  $\phi$  is orientation preserving,  $\phi^* \mathcal{O}_E = \mathcal{O}_E$ , from which it follows that

$$\sigma_1^* \mathcal{O}_E = (\phi \circ \sigma)^* \mathcal{O}_E = \sigma^* (\phi^* \mathcal{O}_E) = \sigma^* \mathcal{O}_E,$$

that is,  $\sigma$  and  $\sigma_1$  induce the same orientation on  $U_p$ . Thus, we can define a global orientation on  $M$  by defining on each evenly covered open subset of  $M$ , the pullback orientation induced by any local section over it. By the above discussion it is clear that such an assignment of orientation on  $M$  is well defined, as the orientations on any two overlapping evenly covered open subsets will always agree.  $\square$

**Lemma A.3.0.6. (Characteristic property of the orientation covering).** *Let  $M$  be a non-orientable smooth manifold and let  $\hat{\pi} : \hat{M} \rightarrow M$  be its orientation covering. If  $X$  is any orientable smooth manifold with or without boundary and  $F : X \rightarrow M$  is any local diffeomorphism, then, there exists a unique orientation preserving local diffeomorphism  $\hat{F} : X \rightarrow \hat{M}$  such that the following diagram commutes.*

$$\begin{array}{ccc} & & \hat{M} \\ & \nearrow \hat{F} & \downarrow \hat{\pi} \\ X & \xrightarrow{F} & M \end{array}$$

**Theorem A.3.0.7. (Uniqueness of the orientation covering).** *Let  $M$  be a non-orientable smooth manifold and let  $\hat{\pi} : \hat{M} \rightarrow M$  be its orientation covering. If  $\tilde{M}$  is an oriented smooth manifold with or without boundary that admits a two-sheeted covering map  $\tilde{M} \rightarrow M$ , then there exists orientation preserving diffeomorphism  $\phi : \tilde{M} \rightarrow \hat{M}$  such that the following diagram commutes.*

$$\begin{array}{ccc} & & \hat{M} \\ & \nearrow \phi & \downarrow \hat{\pi} \\ \tilde{M} & \xrightarrow{\tilde{\pi}} & M \end{array}$$